

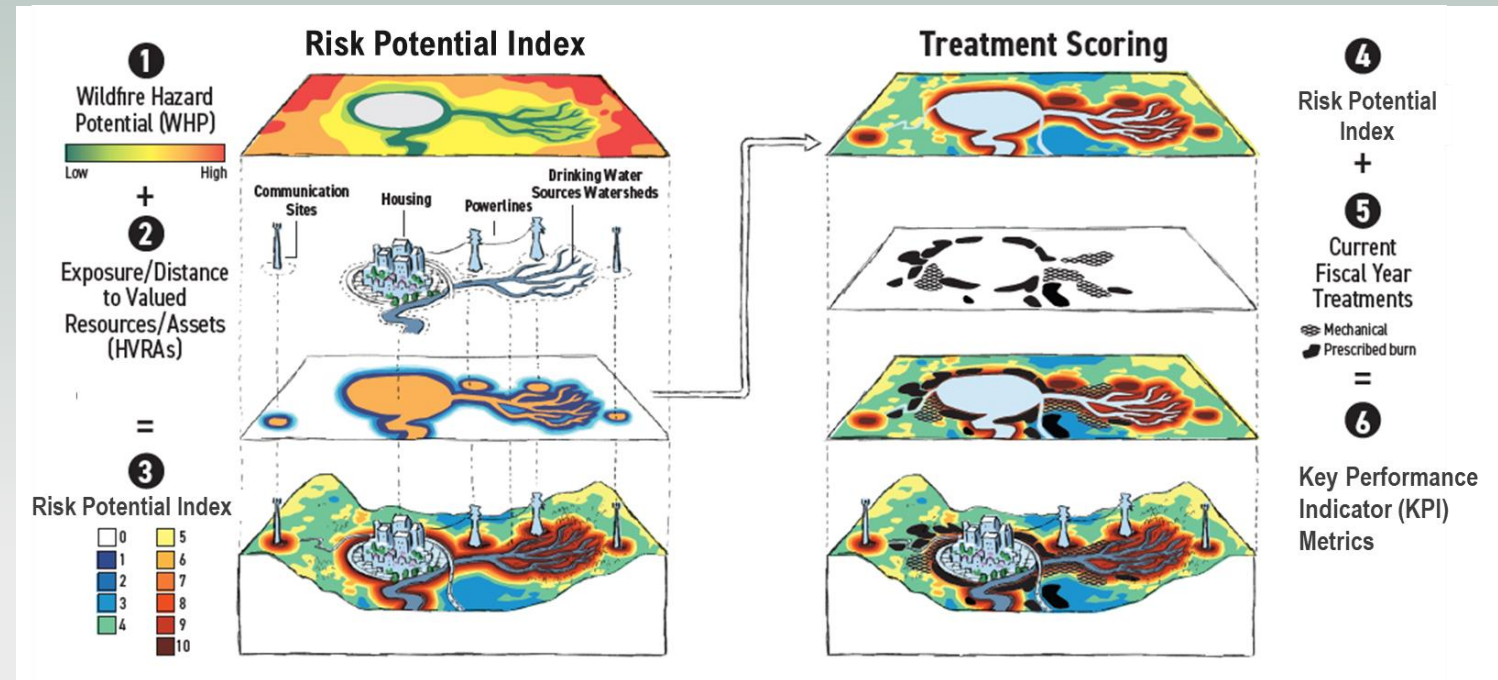


Forest Service
U.S. DEPARTMENT OF AGRICULTURE

Wildfire Risk Reduction Key Performance Indicator

FS-DOI Risk Map Group
February 18, 2026

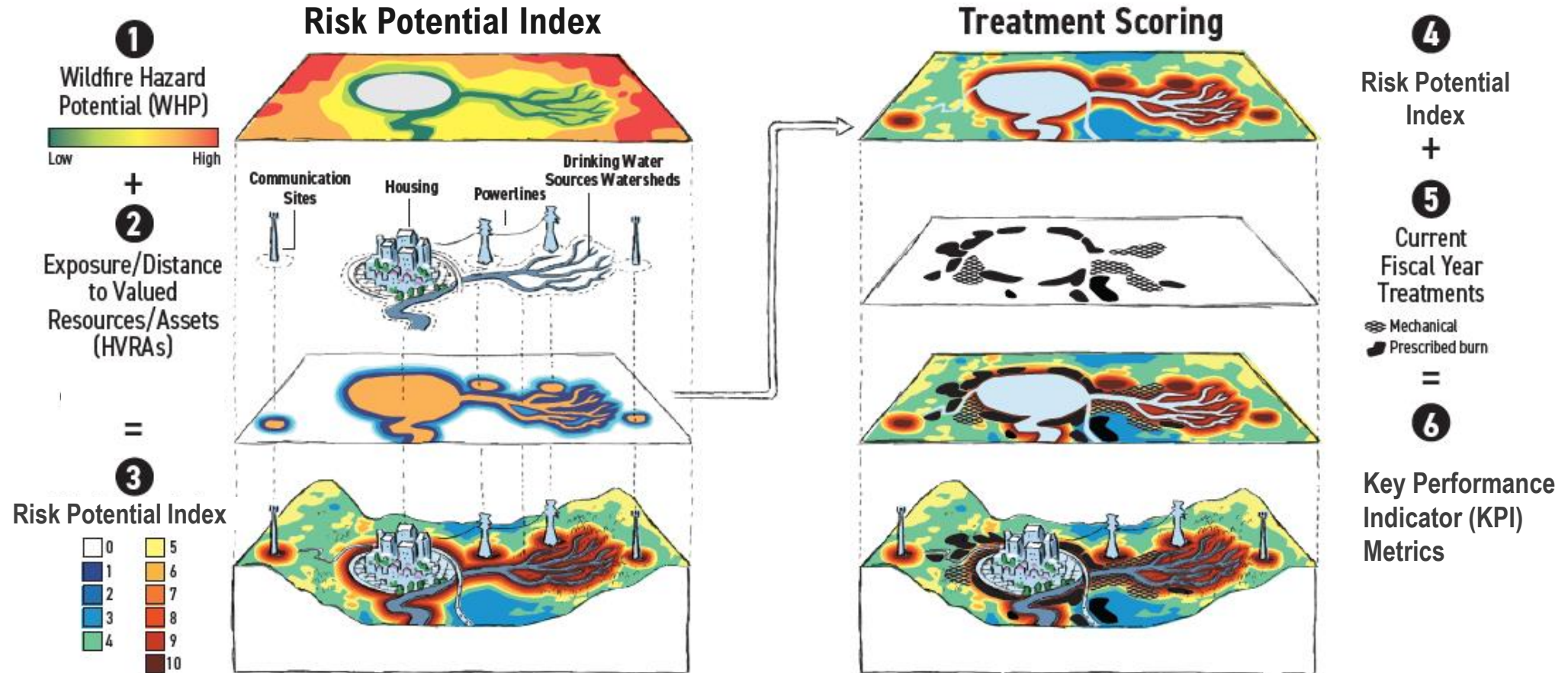
*WO FAM Fuels, Fire Ecology and Smoke
&
RMRS Fire Modeling Institute*



Outline

- What is the WRR-KPI? What is it NOT?
- WRR-KPI Framework overview
- WRR-KPI Framework Step-Through
 - Risk Potential Index Mapping
 - Treatment Condition and Rotation
 - Annual Accounting
 - Summaries and Metrics
- Questions

How do we identify risk “potential”? How are treatments evaluated?



How does this relate to other efforts?

Quantitative Wildfire Risk Assessment (QWRA)

Where is wildfire risk across the Nation, Region, Forest, District?

- Fire behavior modeled and risk calculated for a snapshot in time
- Requires prioritization of assets

Outcome-Based Change Metrics (OBCM)

How has wildfire risk changed as a result of treatments and disturbances?

- Fire behavior modeled, and risk calculated for pre- and post-disturbance conditions
- Limited to 21 “High-Risk” landscapes, for now

Wildfire Risk Reduction Key Performance Indicator (WRR-KPI)

As an agency, are we putting fuels treatments in places that will reduce wildfire risk?

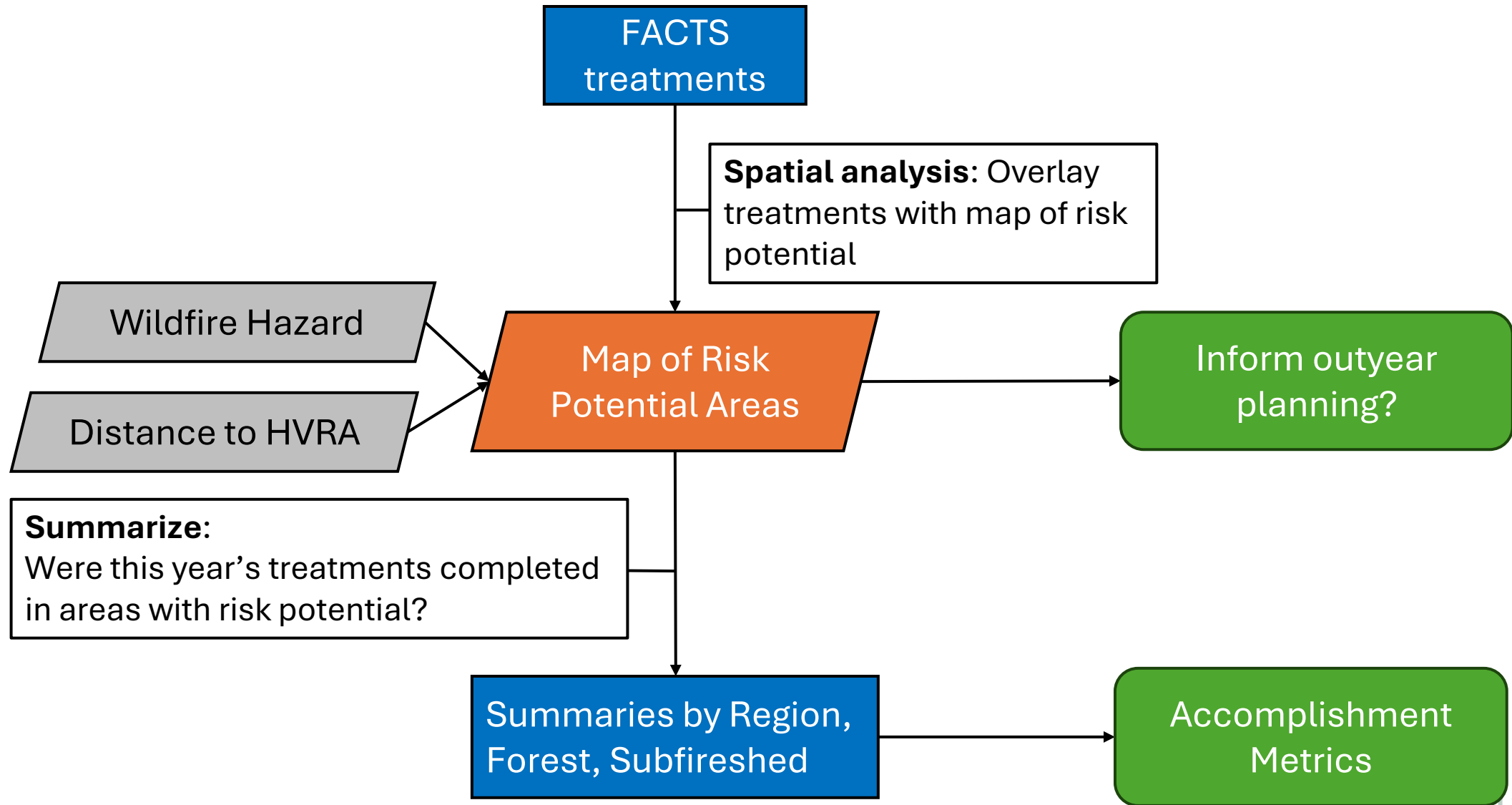
- Annual treatments compared against assessment of wildfire hazard and distance to HVRAs
- Assessed for all NFS

KPI “Feasibility Study”

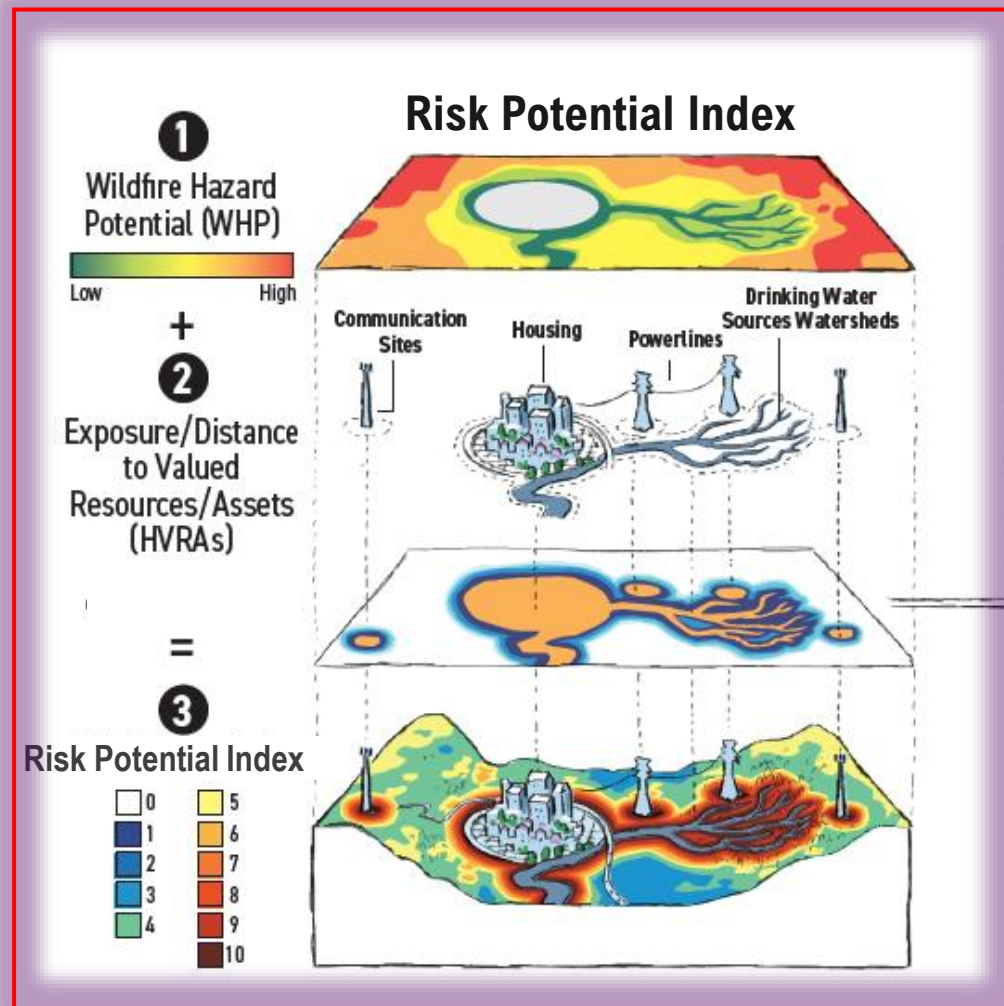
How can we optimize assessment of agency accomplishments?

- Optimize fire behavior modeling?
- Develop a statistical surrogate?

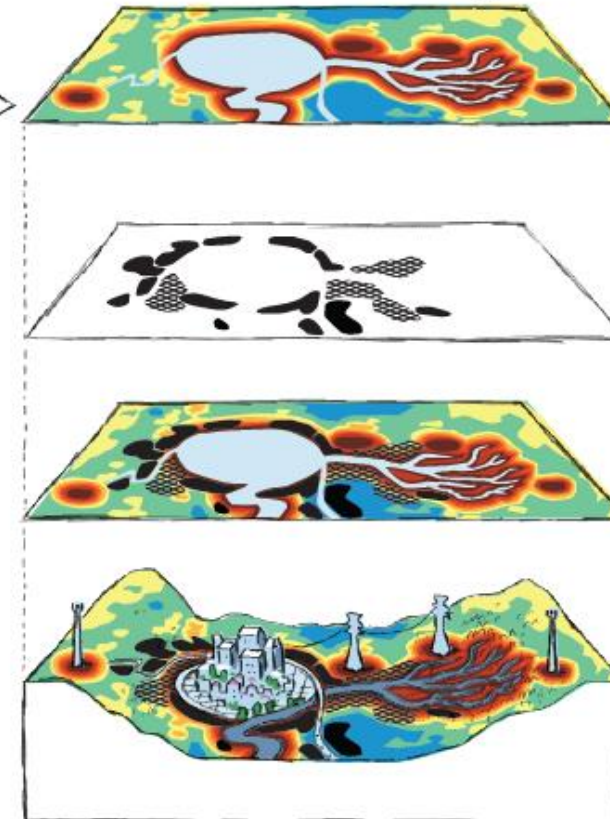
WRR-KPI Process Overview



Risk Potential Index Mapping



Treatment Scoring



4
Risk Potential Index
+

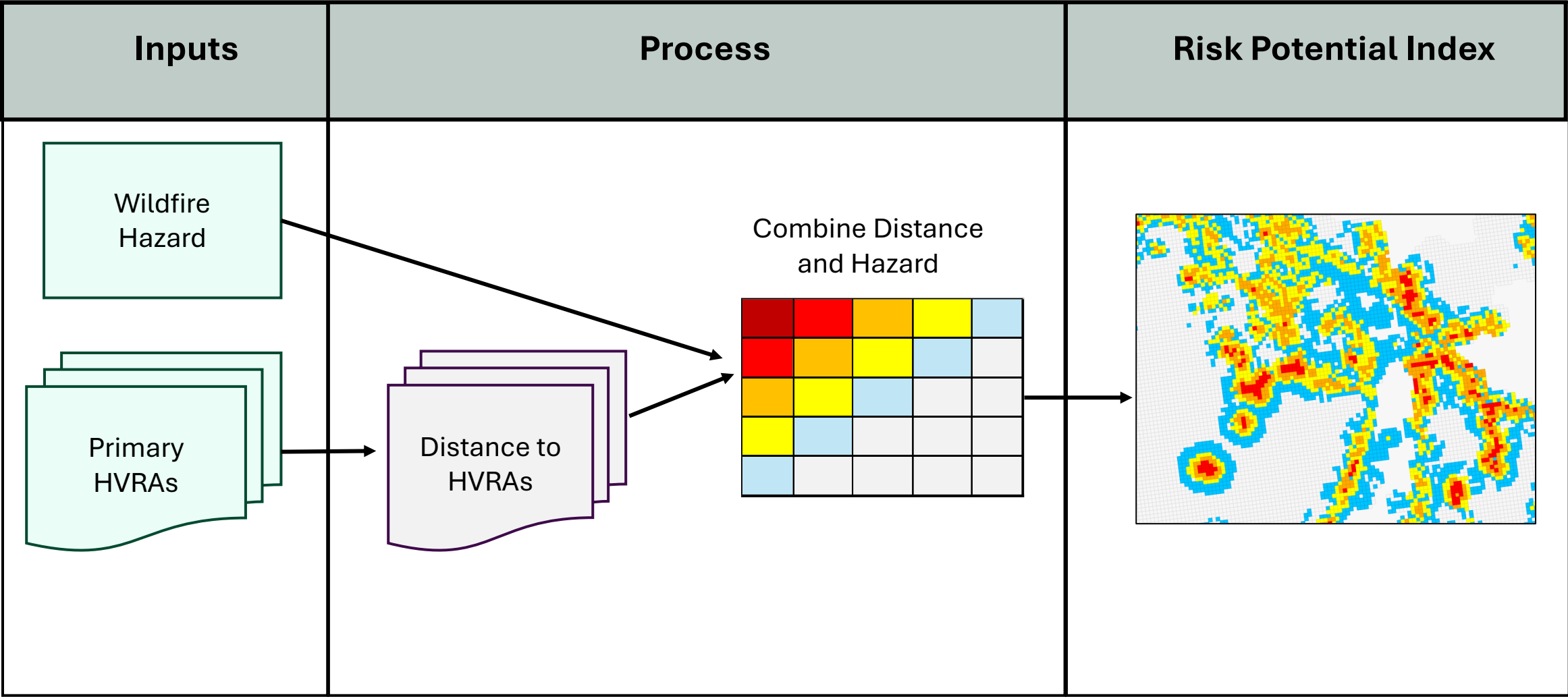
5
Current Fiscal Year Treatments

Mechanical
Prescribed burn

=

6
Key Performance Indicator (KPI) Metrics

Risk Potential Index Classification



Input Data – first choice is regional

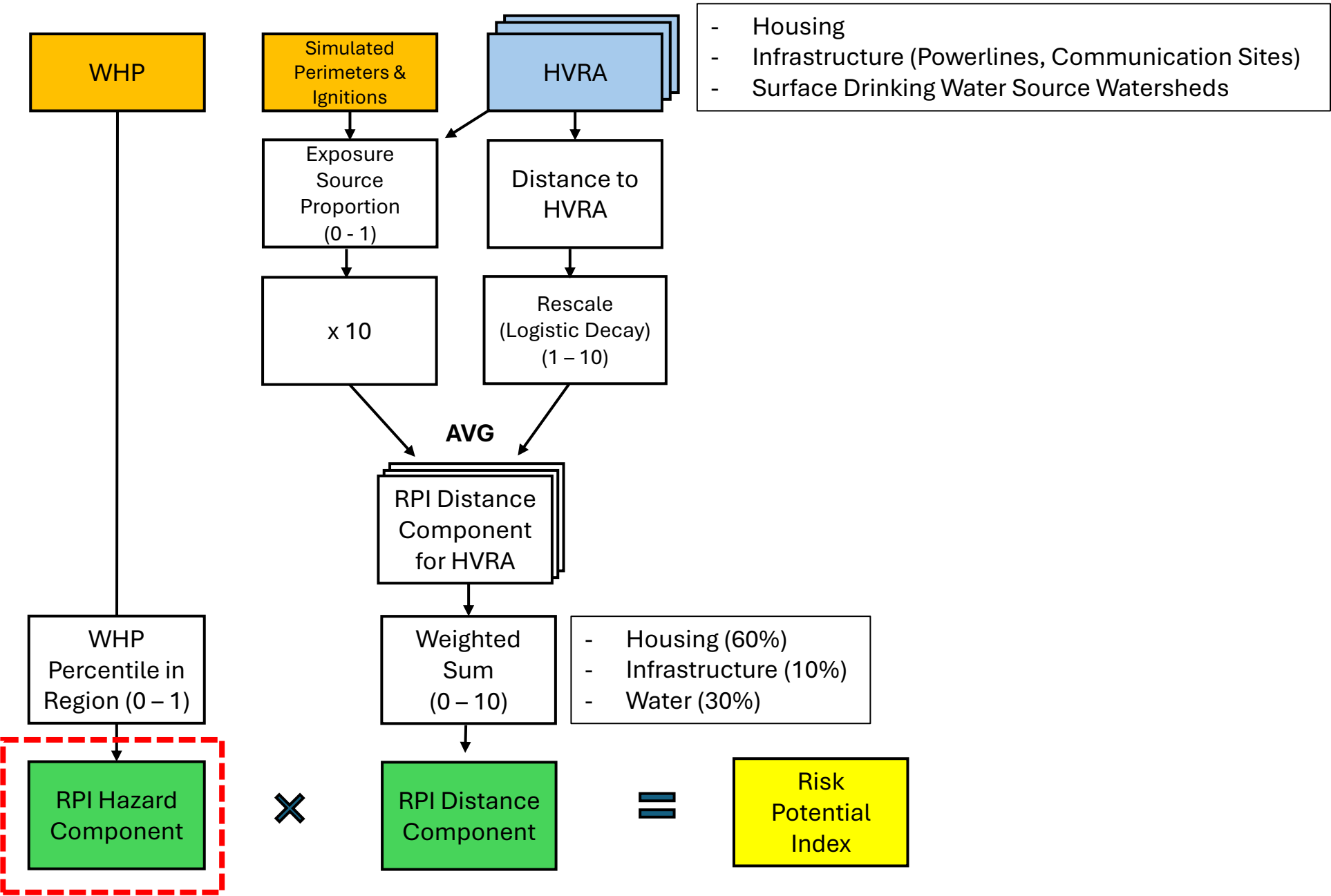
Wildfire Hazard Potential (WHP)

- National: WHP 2023

Highly Valued Resources and Assets (HVRAs)

- *Housing*
 - National: Wildfire Risk to Communities Housing Unit Density
- *Infrastructure (Transmission Lines and Communication Sites)*
 - National: 2022 RMA Dashboard (HIFLD)
- *Drinking Water Supply*
 - National: Recent FMI product based on selected HUCs

Risk Potential
Index
Calculation

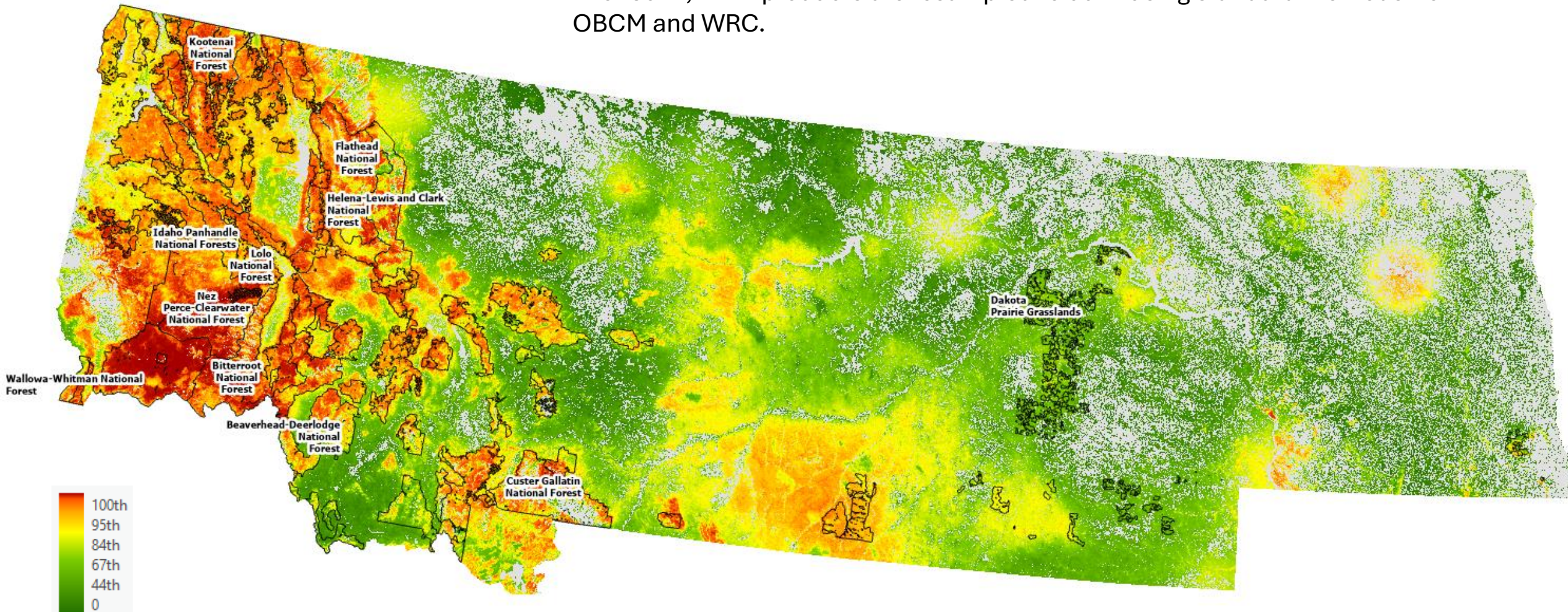


Risk Potential Index Calculation

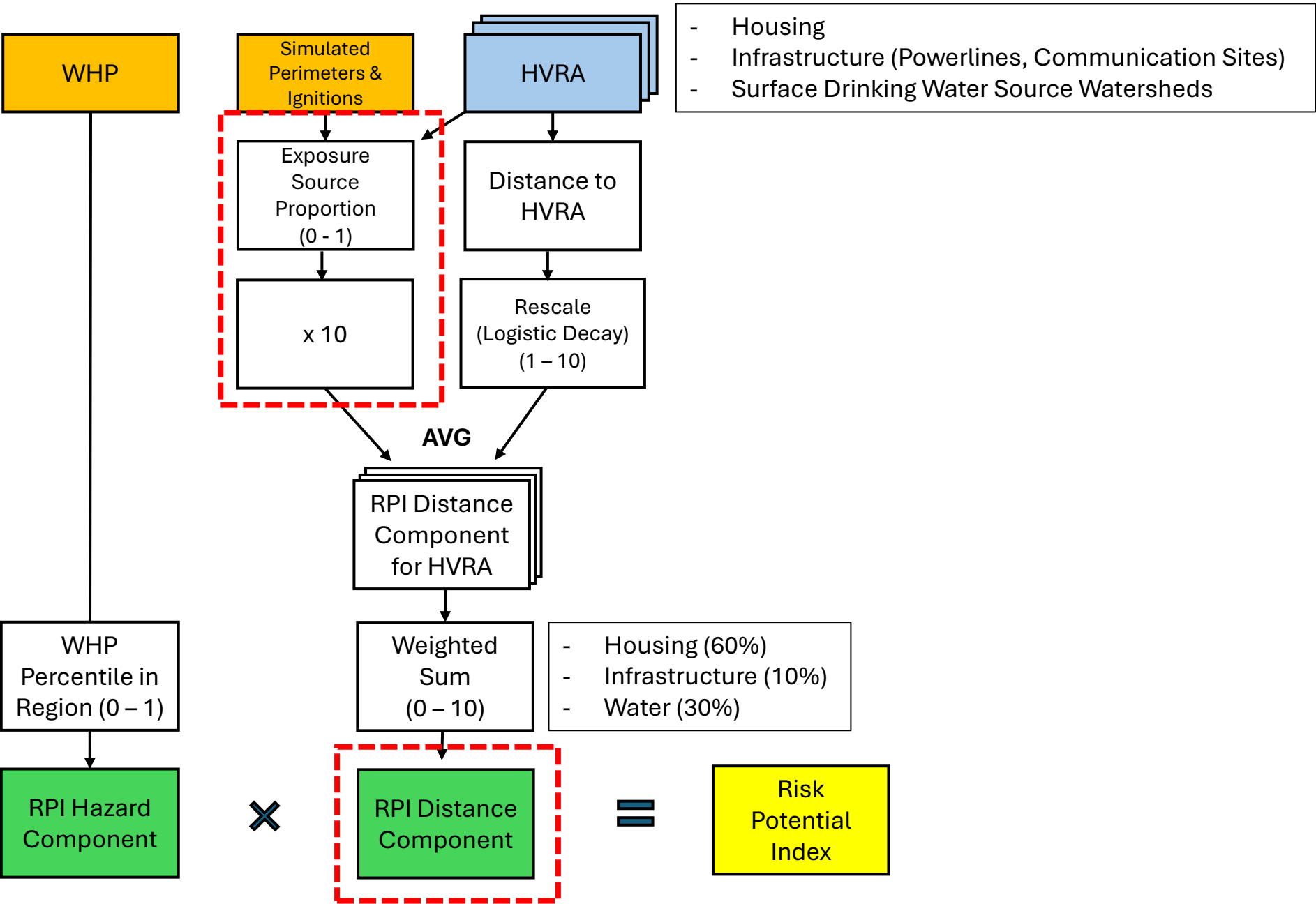
RPI Hazard Component

Wildfire Hazard Potential is converted to a percentile relative to values within the Administrative Regional Boundary.

Because RPI is mapped at 90 m resolution and WHP tends to be calculated at 270 m or 30 m, WHP products are resampled to 90 m using standard methods from OBCM and WRC.



Risk Potential Index Calculation



Risk Potential Index Calculation

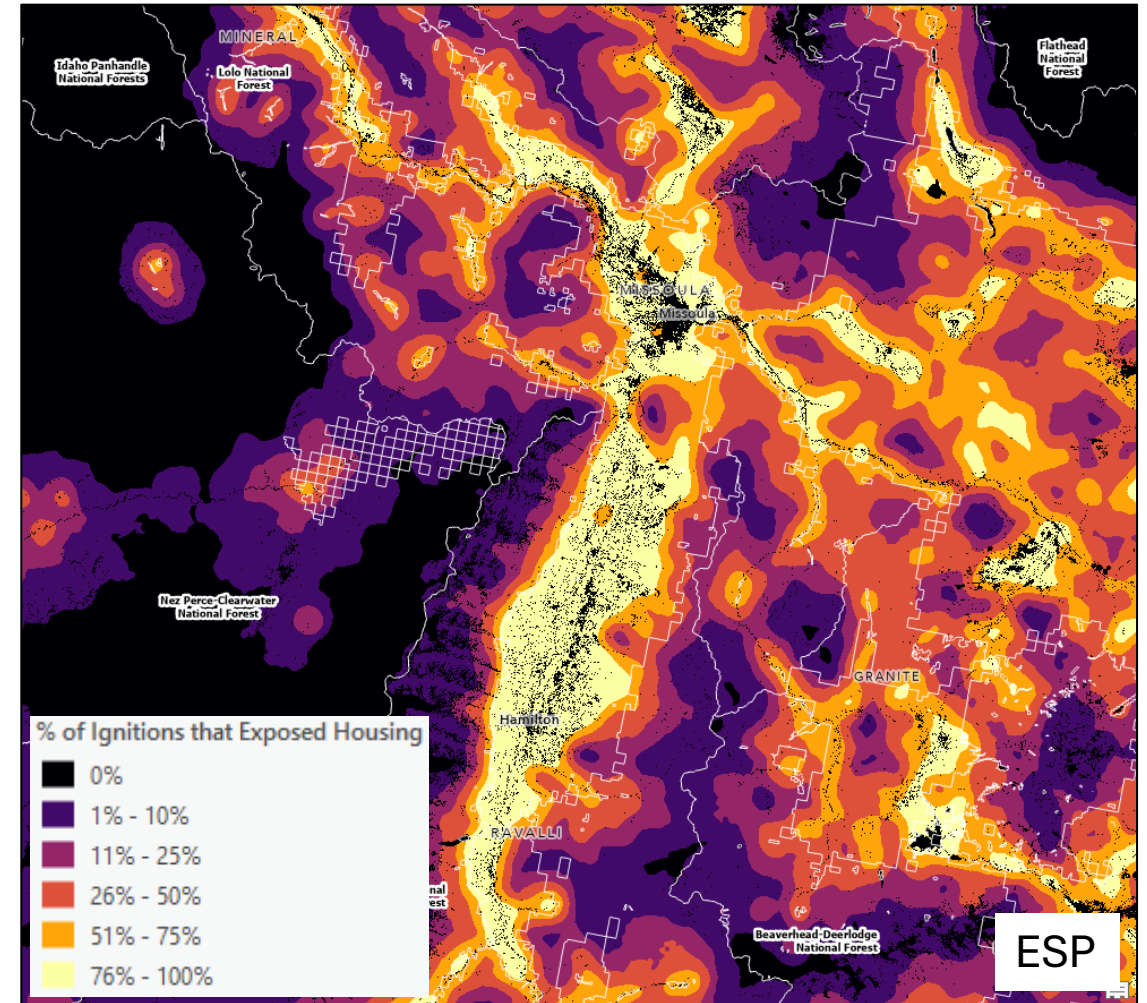
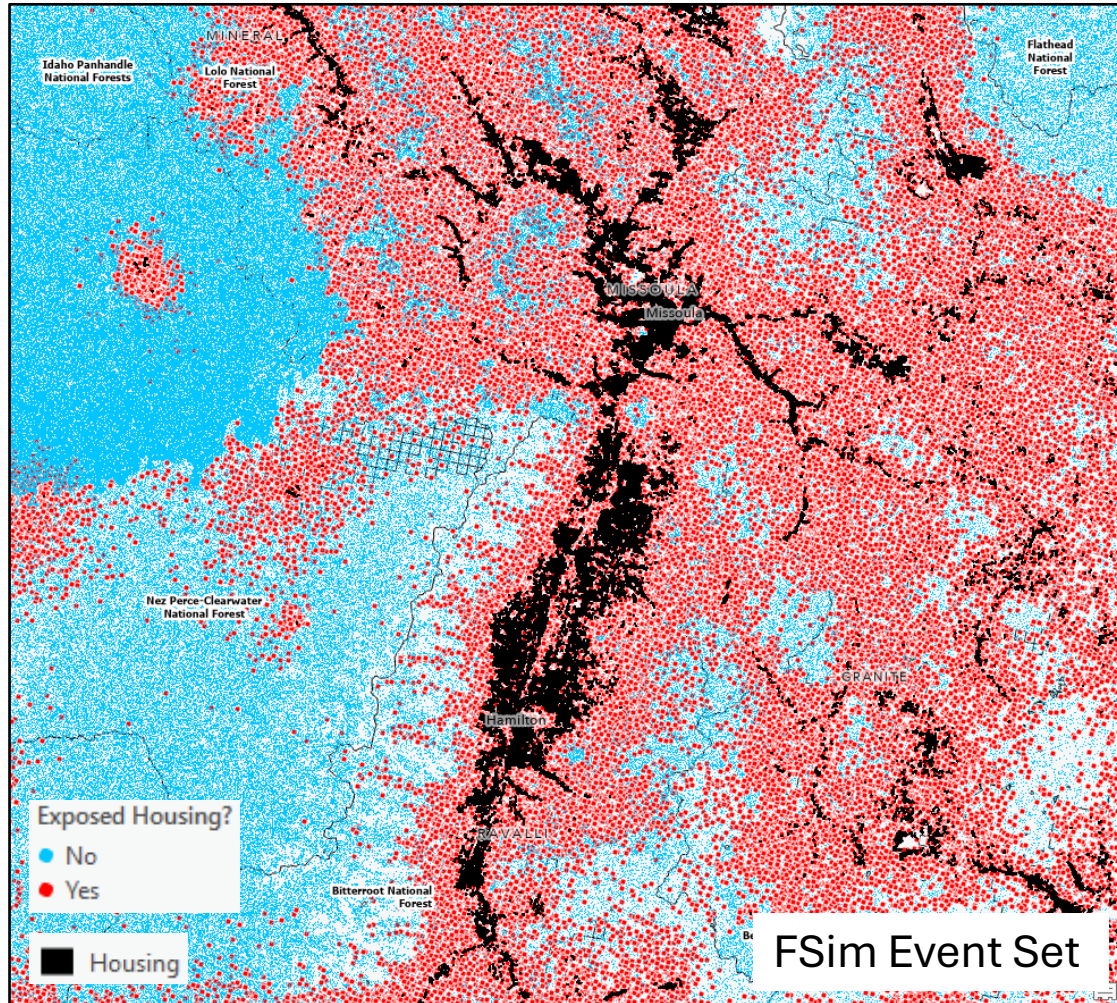
RPI Distance
Component

Exposure
Source
Proportion

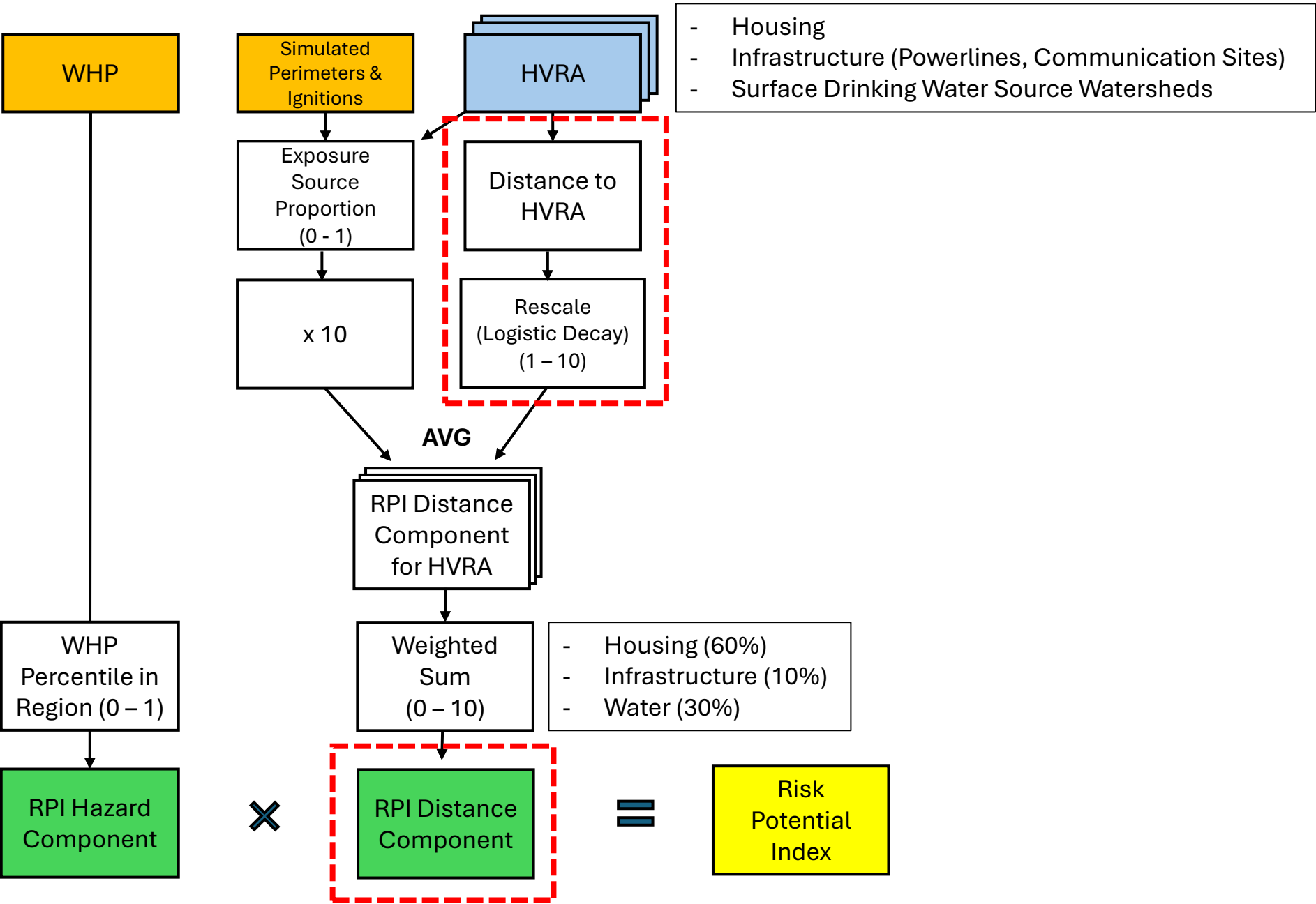
Exposure Source Proportion (ESP) represents the proportion of simulated ignition points of fires that ultimately expose HVRAs.

ESP defines the maximum distance in which treatments could be a barrier to wildfire transmission and a gradient in which treatments could be an effective barrier.

ESP is created for each HVRA.



Risk Potential Index Calculation



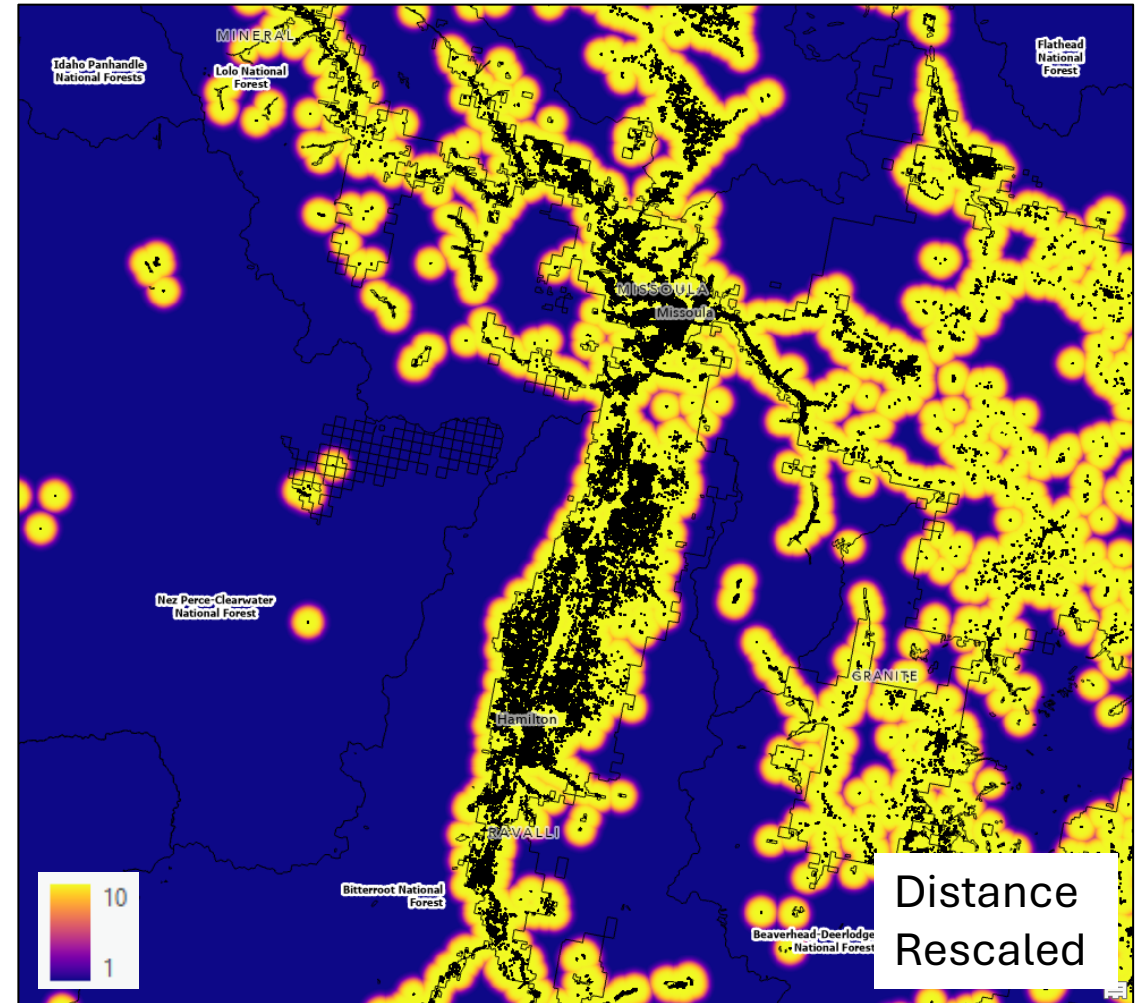
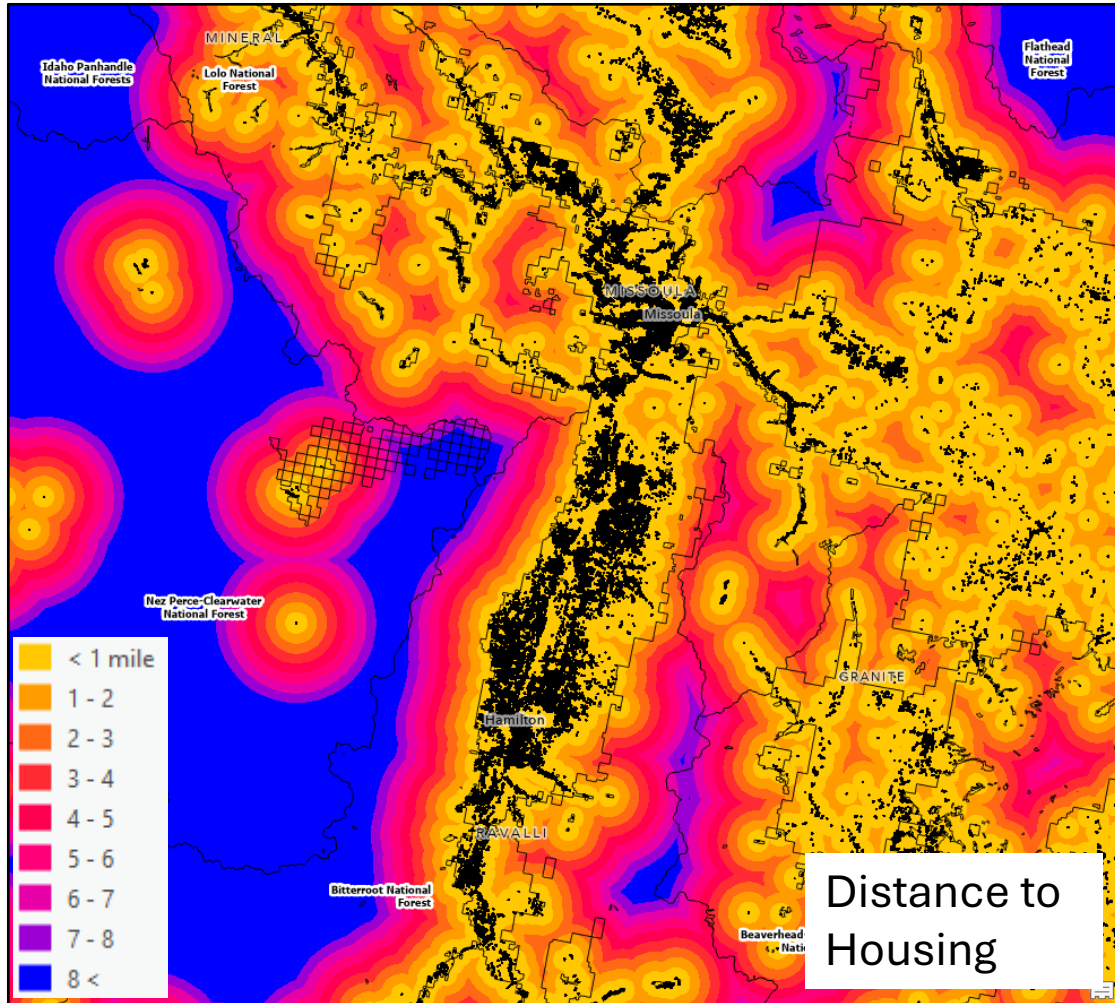
Risk Potential Index Calculation

RPI Distance
Component

Distance to
HVRA

The distance to each HVRA is calculated. The distances are rescaled to a 1 to 10 scale where closer distances have greater value.

A logistic decay function is fit to give the greatest value to treatments placed within 1.5 miles of HVRA. This distance is consistent with the WRC Community Wildfire Risk Reduction Zones and ember transport distances.



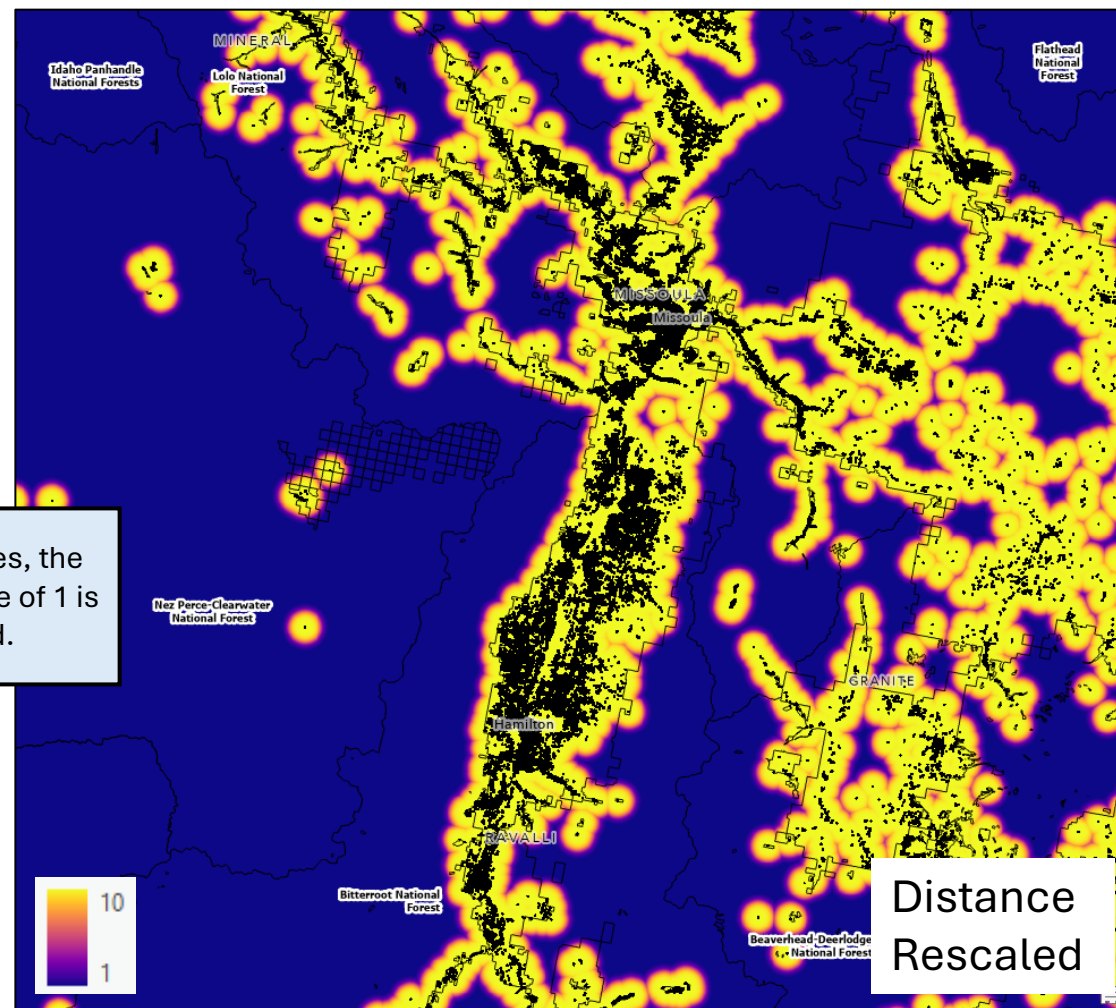
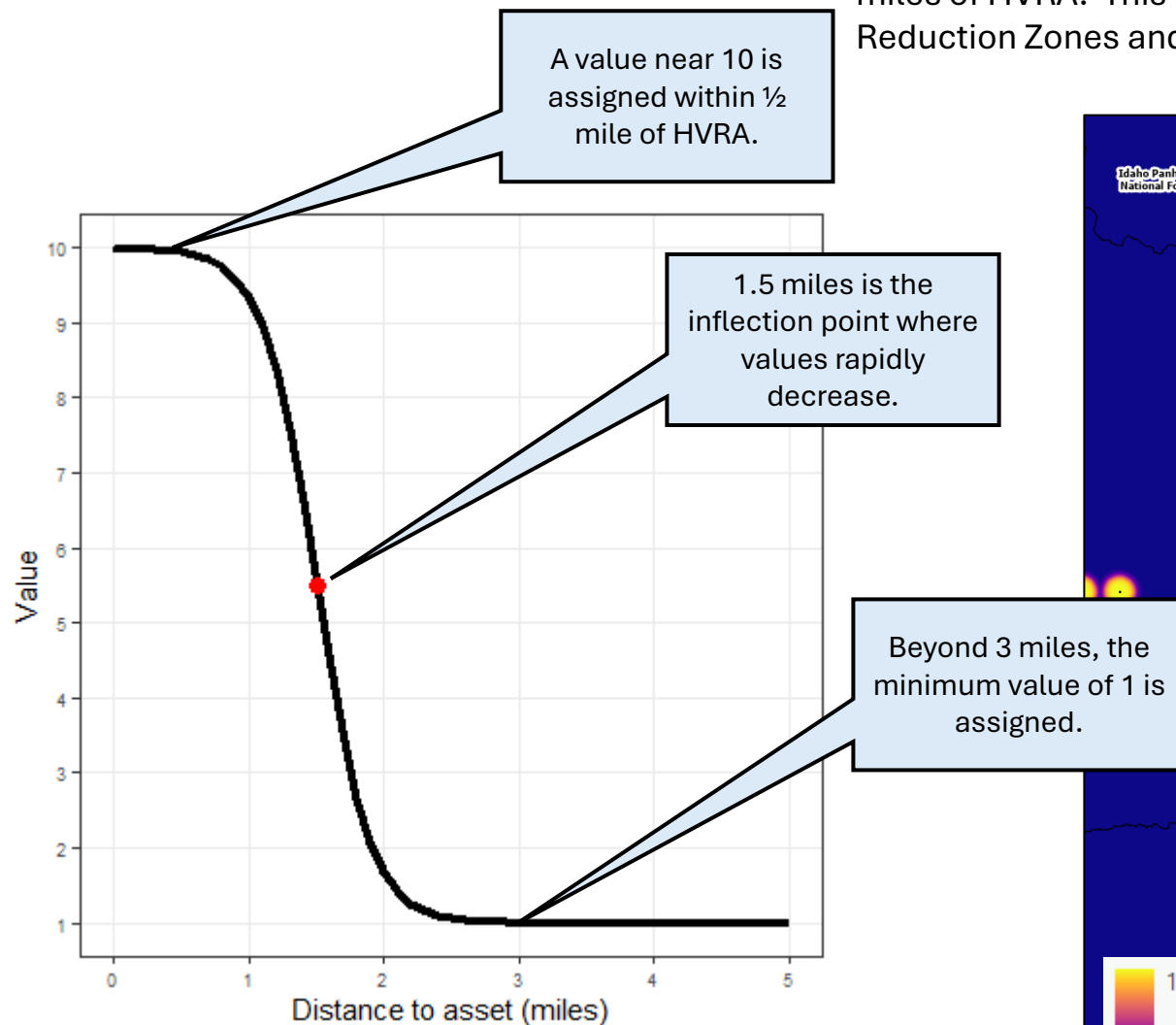
Risk Potential Index Calculation

RPI Distance
Component

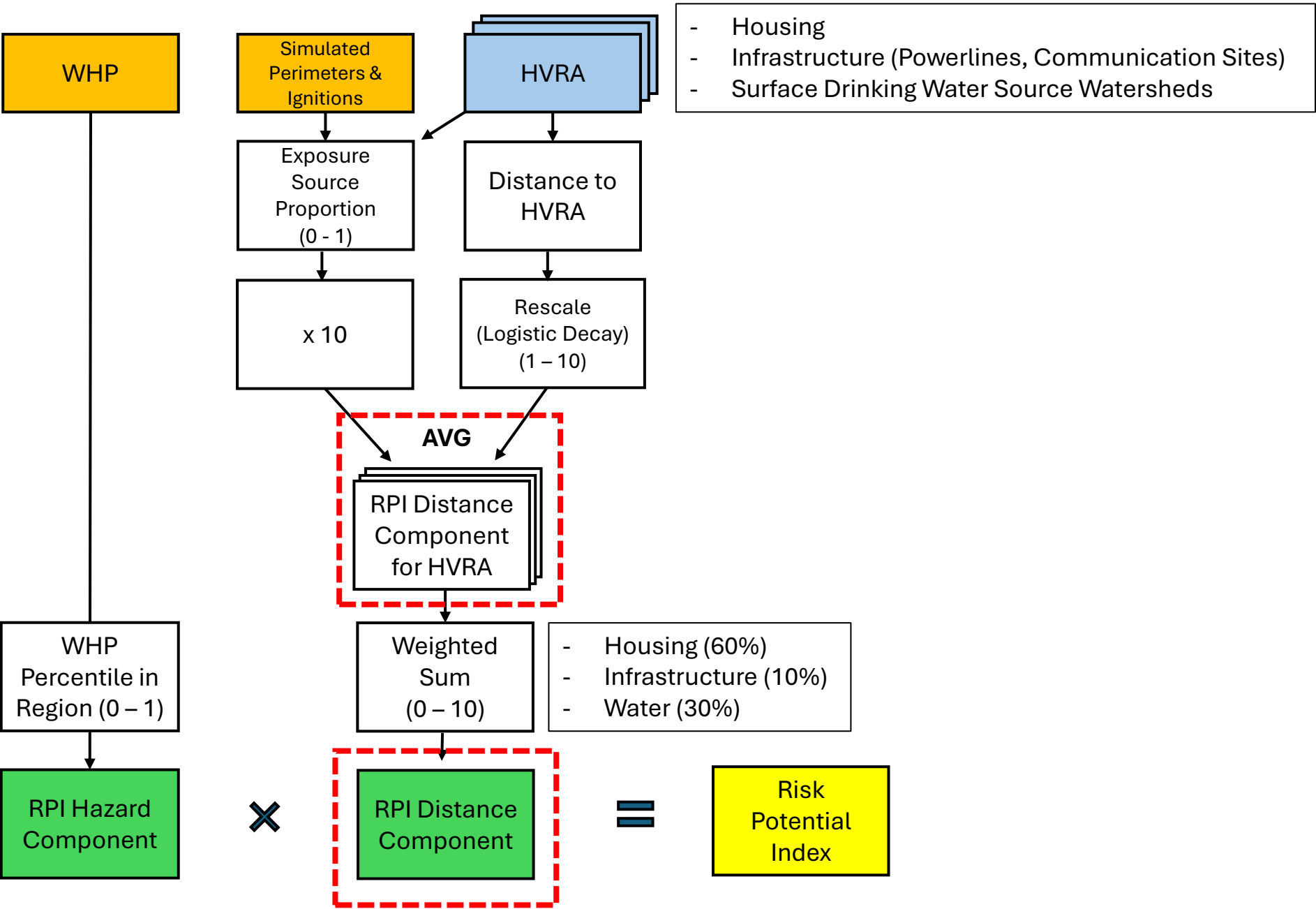
Distance to
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The distance to each HVRA is calculated. The distances are rescaled to a 1 to 10 scale where closer distances have greater value.

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Risk Potential
Index
Calculation

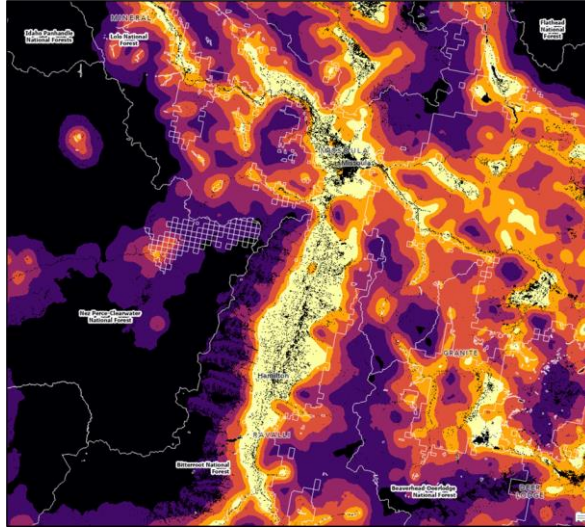


Risk Potential
Index
Calculation

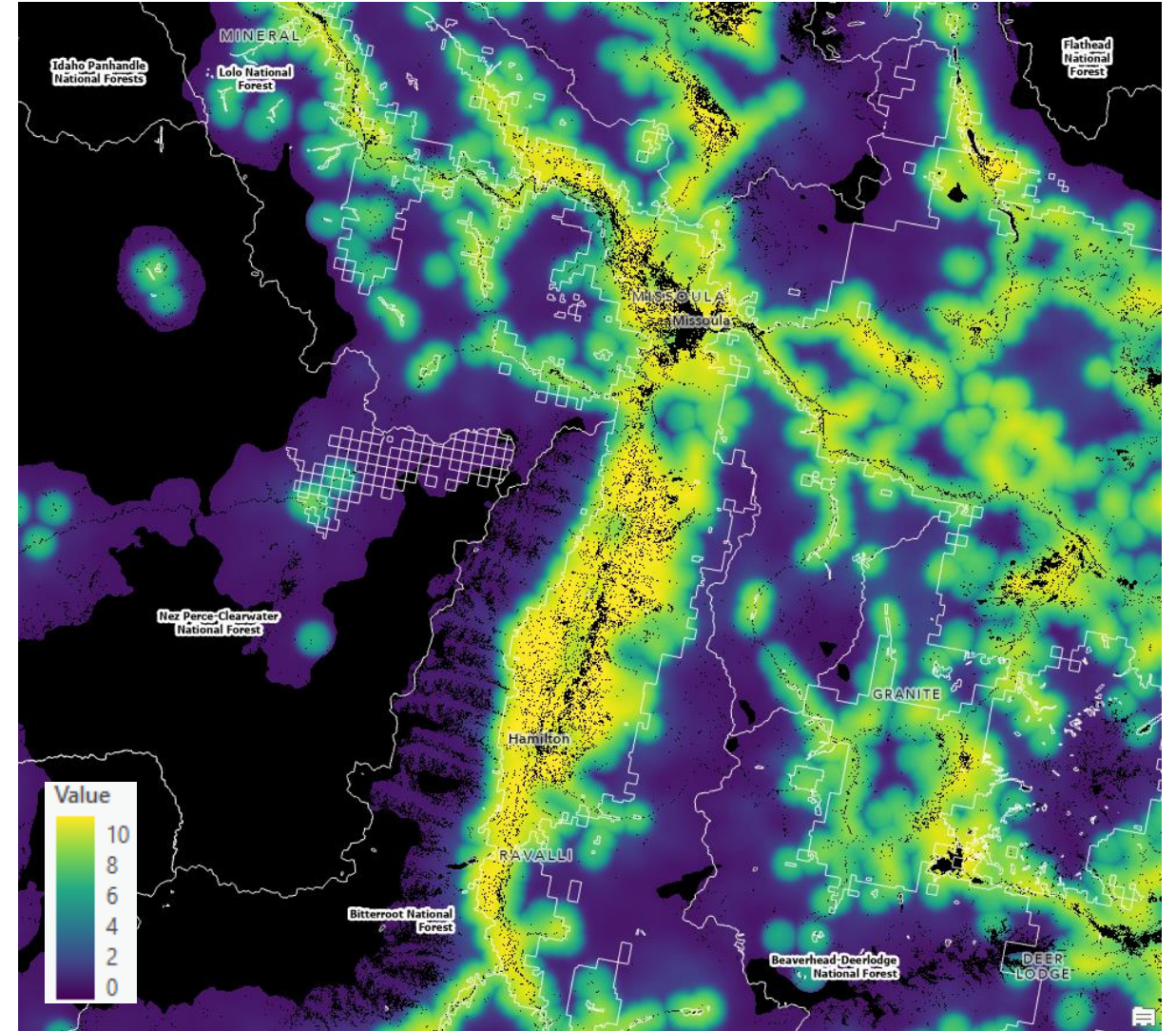
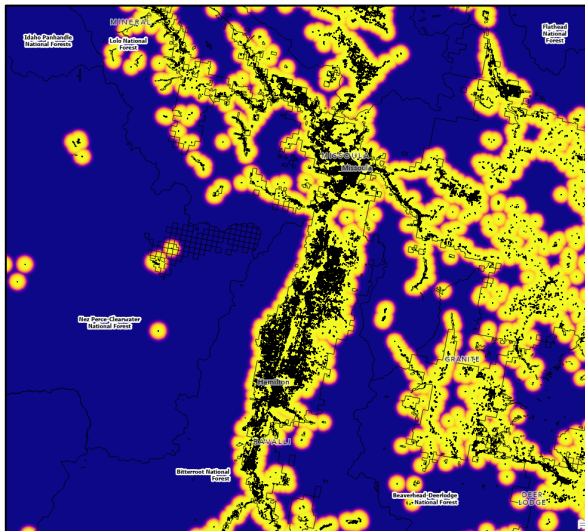
RPI
Distance
Component

The ESP and Rescaled Distance are combined to
create the RPI Distance Component for each HVRA.

Exposure
Source
Proportion
(ESP)



Distance
to HVRA



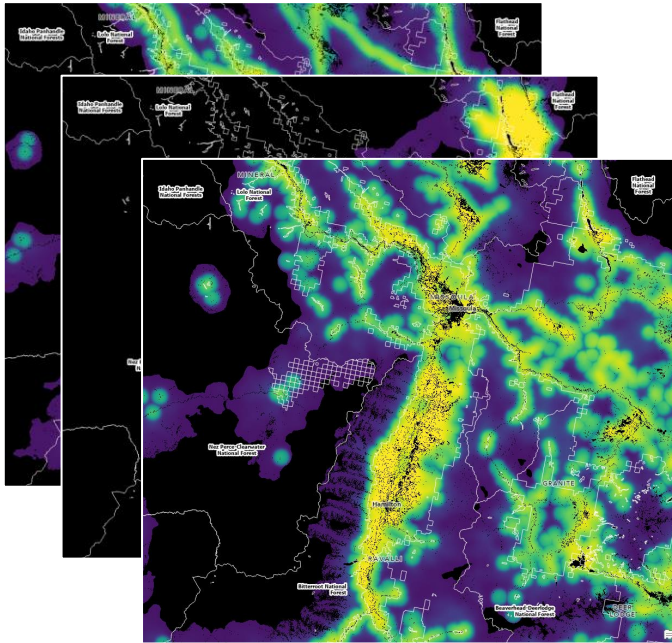
Risk Potential Index Calculation

RPI Distance
Component

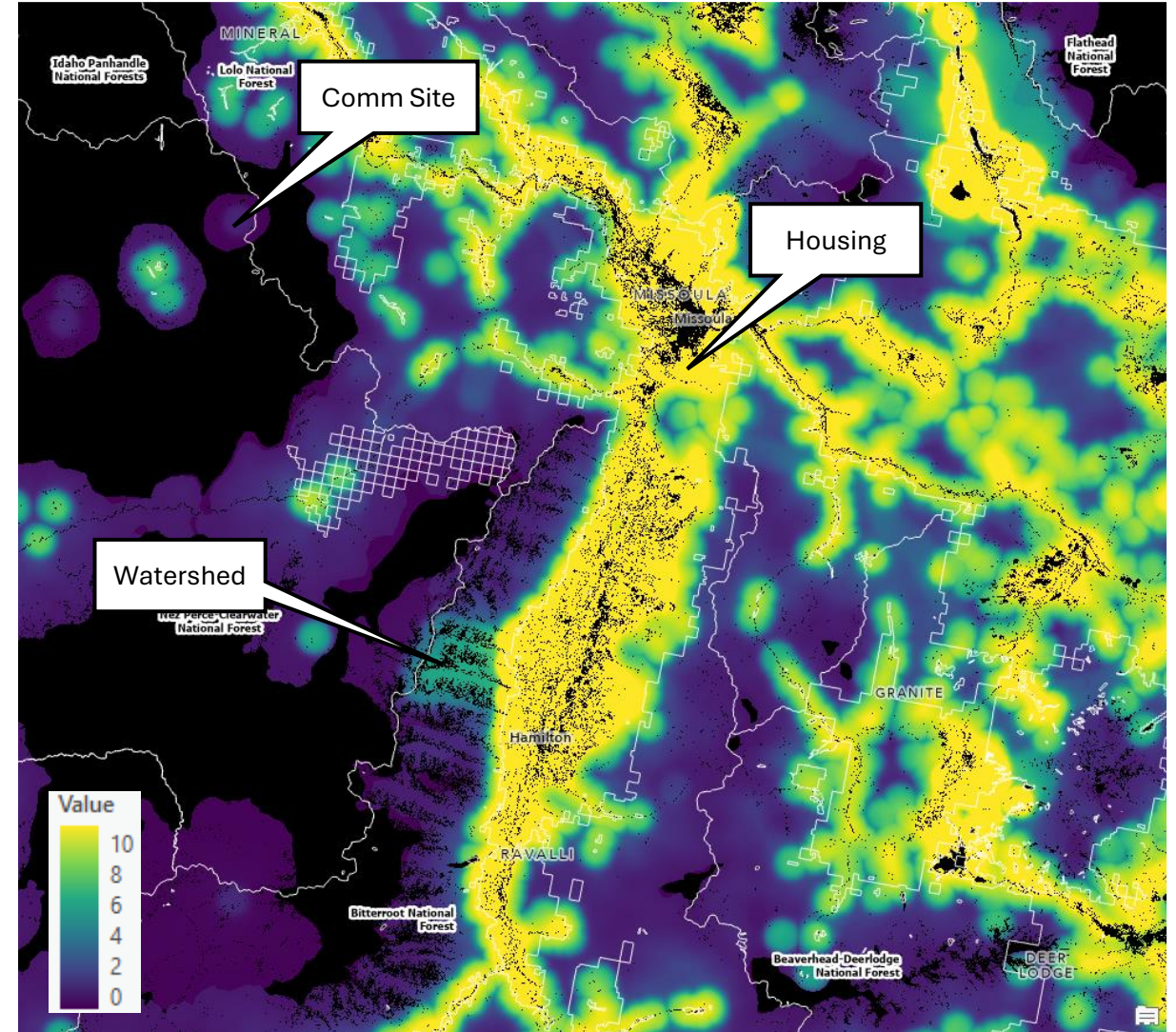
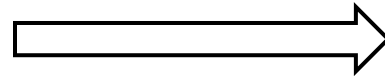
The RPI Distance Components for all HVRA are combined using weights consistent with the National QWRA:

$$(\text{Housing} \times 0.6) + (\text{Surface Drinking Water} \times 0.3) + (\text{Infrastructure} \times 0.1)$$

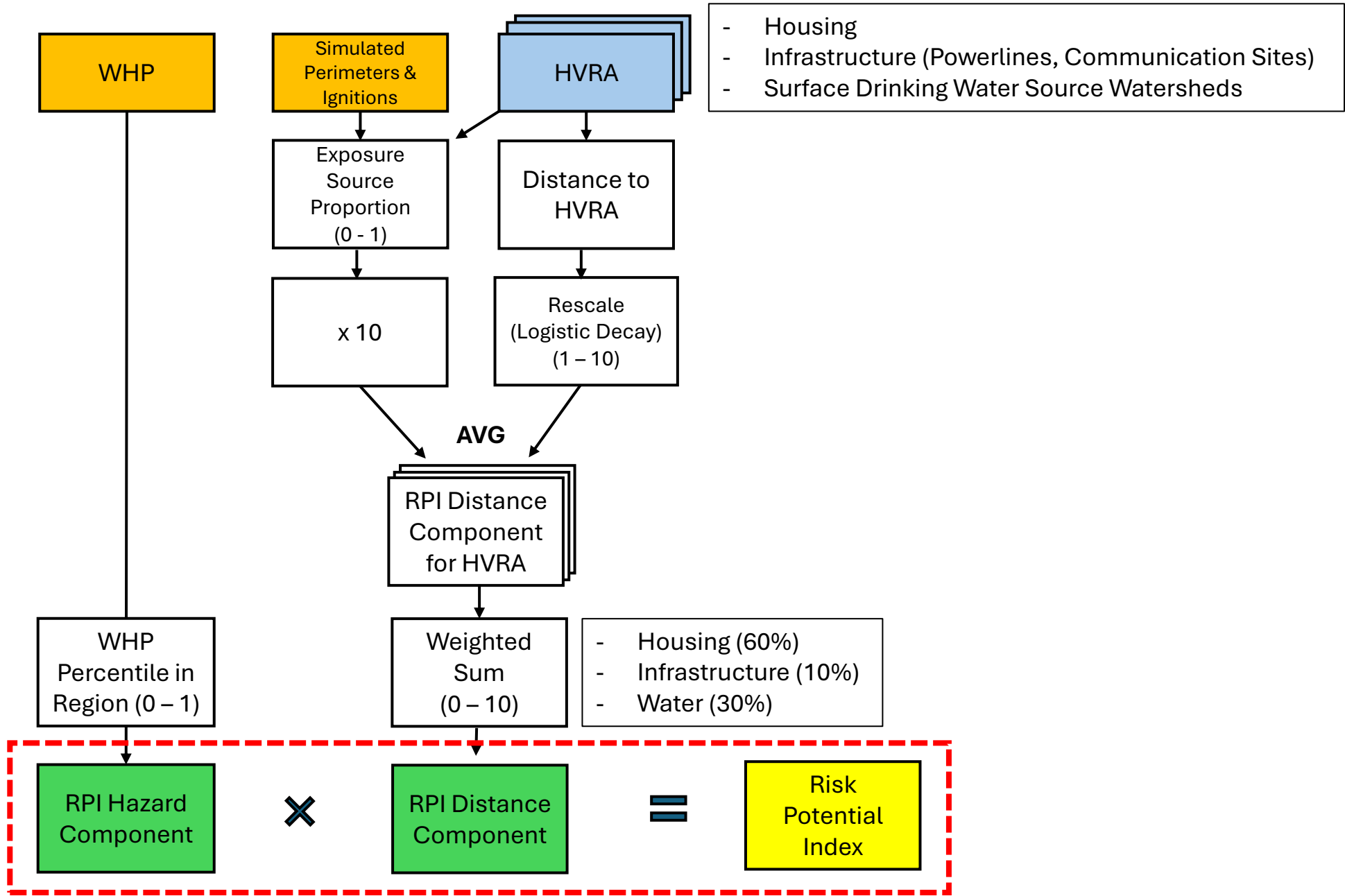
RPI Distance Component for each HVRA



Housing: 60%
Surface Drinking Water: 30%
Infrastructure: 10%



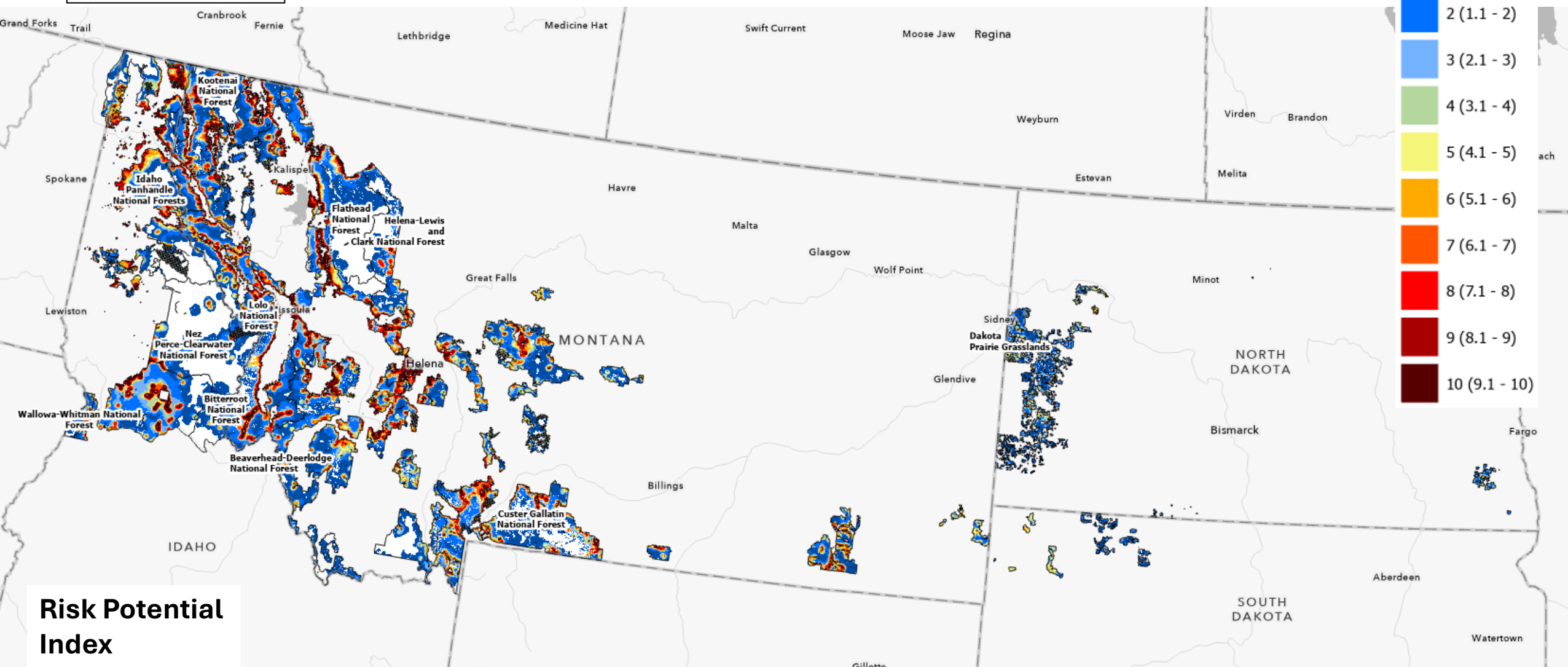
Risk Potential
Index
Calculation



Risk Potential Index Calculation

Risk Potential Index

The RPI Hazard and Distance Components are combined to create the final Risk Potential Index.

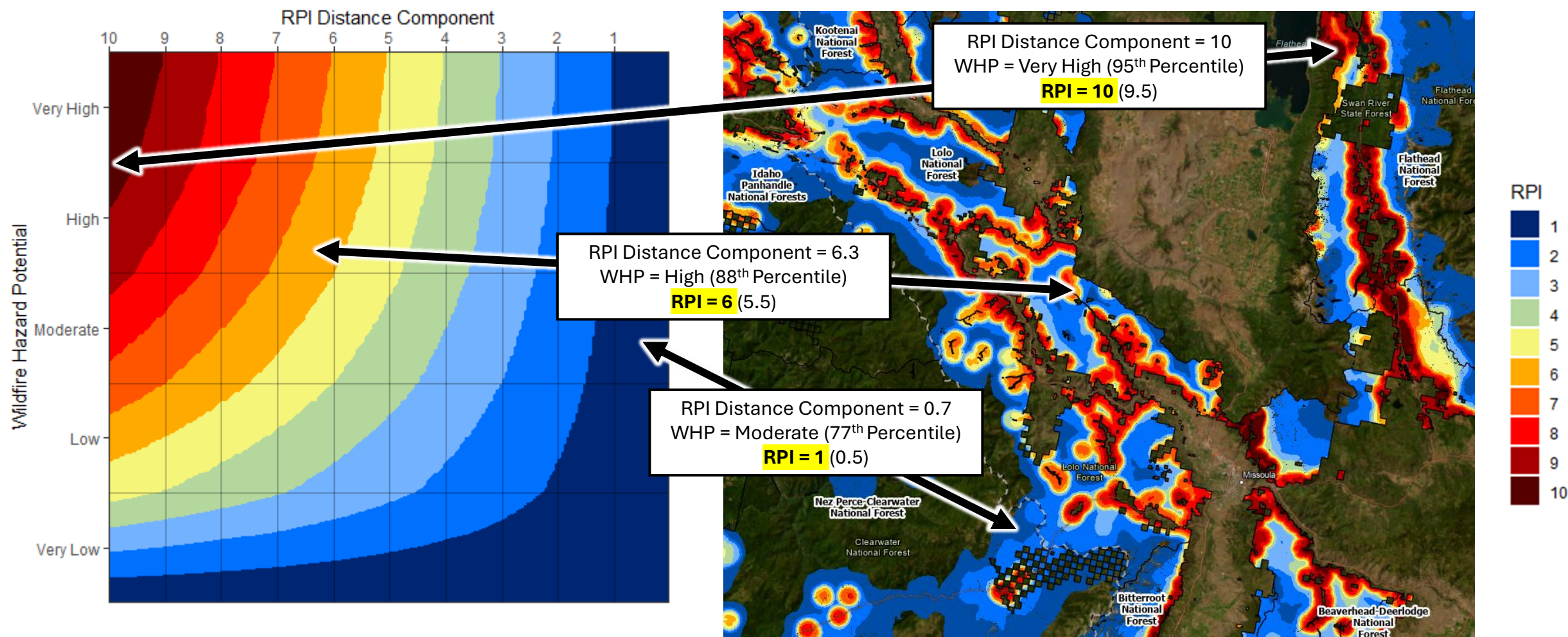


Risk Potential Index

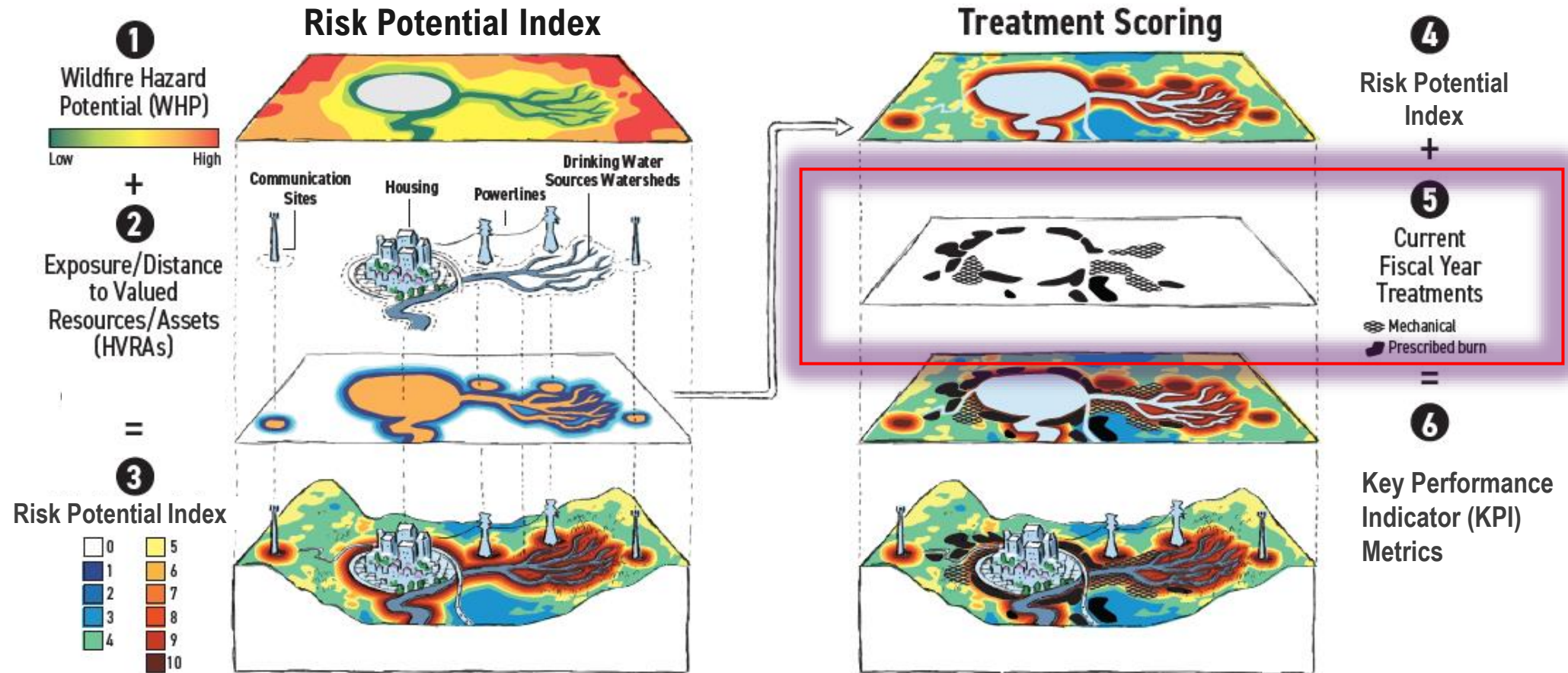
Risk Potential
Index
Calculation

Risk
Potential
Index

$$\text{RPI Distance Component} \times \text{WHP Percentile} = \text{Final RPI}$$



Fuels Treatment Data

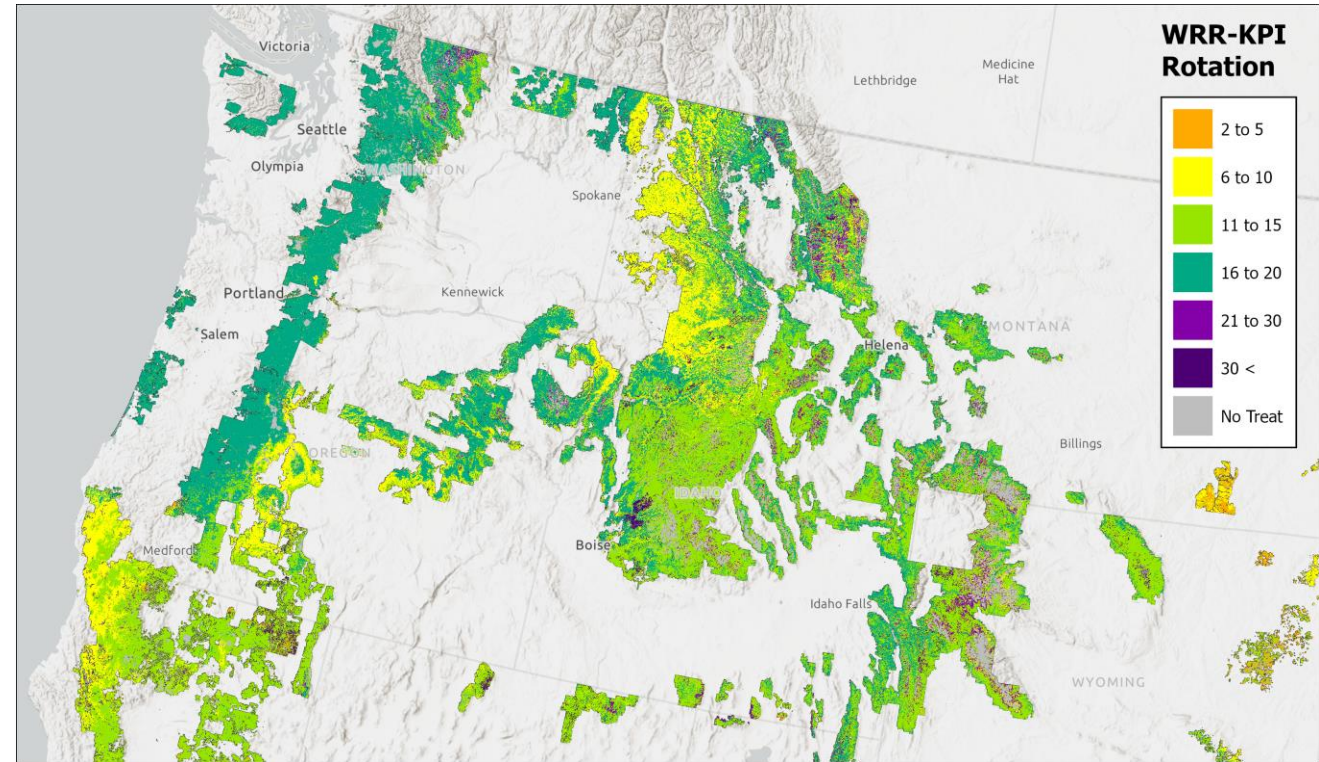


WRR-KPI Rotation

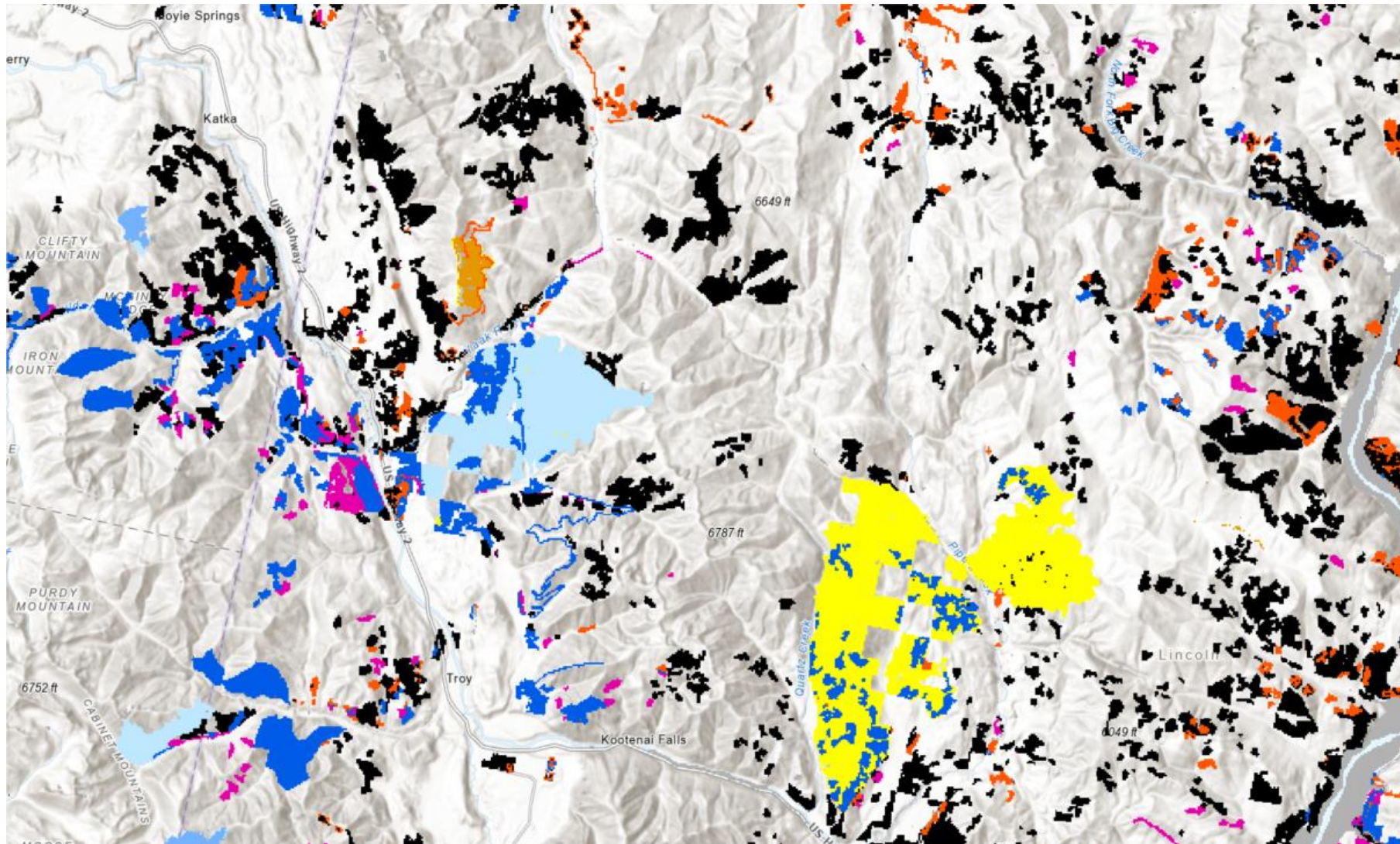
- Treatment Condition is based on:
 - Type of treatment (mitigation or intermediate)
 - Time since last treatment
 - Rotation values for each vegetation type
- WRR-KPI Rotation is hazard-based, not ecologically-based ($\neq$ Mean Fire Return Interval)
- The rotation selected for each Existing Vegetation Type directly impacts the annual KPI accounting

WRR-KPI Rotation

EVT_NAME	R1	R2	R3	R4	R5	R6	R8	R9
Rocky Mountain Subalpine Dry-Mesic Spruce-Fir Forest and Woodland	11 to 15	11 to 15	11 to 15	11 to 15	11 to 15	11 to 15		
Rocky Mountain Lodgepole Pine Forest	11 to 15	11 to 15		11 to 15		11 to 15		
Inter-Mountain Basins Montane Sagebrush Steppe	11 to 15	11 to 15	11 to 15	11 to 15	11 to 15	11 to 15		
Northern Rocky Mountain Dry-Mesic Montane Mixed Conifer Forest	16 to 20			16 to 20	16 to 20	16 to 20		
Southern Rocky Mountain Ponderosa Pine Woodland		6 to 10	6 to 10	6 to 10				
Northern Rocky Mountain Mesic Montane Mixed Conifer Forest	6 to 10			16 to 20		16 to 20		
Middle Rocky Mountain Montane Douglas-fir Forest and Woodland	16 to 20	16 to 20		16 to 20				
Mediterranean California Mesic Mixed Conifer Forest and Woodland				11 to 15	11 to 15	11 to 15		
Madrean Pinyon-Juniper Woodland			11 to 15					
Colorado Plateau Pinyon-Juniper Woodland		21 to 30	16 to 20	21 to 30				
Rocky Mountain Aspen Forest and Woodland	16 to 20	16 to 20	16 to 20	16 to 20	16 to 20	16 to 20		
Northern Rocky Mountain Ponderosa Pine Woodland and Savanna	6 to 10			6 to 10		11 to 15		
Great Basin Pinyon-Juniper Woodland				21 to 30	6 to 10	21 to 30		
Rocky Mountain Subalpine Mesic-Wet Spruce-Fir Forest and Woodland	16 to 20	16 to 20	16 to 20	16 to 20		16 to 20		
Northern Rocky Mountain Subalpine Woodland and Parkland	21 to 30	21 to 30		21 to 30		21 to 30		
Laurentian-Acadian Northern Hardwoods Forest								6 to 10
Rocky Mountain Cliff Canyon and Massive Bedrock	No Treat	No Treat	No Treat	No Treat		No Treat		
North Pacific Maritime Dry-Mesic Douglas-fir-Western Hemlock Forest					16 to 20	16 to 20		
Rocky Mountain Alpine Bedrock and Scree	No Treat	No Treat	No Treat	No Treat		No Treat		
North Pacific Mesic Western Hemlock-Silver Fir Forest						16 to 20		
Mediterranean California Dry-Mesic Mixed Conifer Forest and Woodland				11 to 15	11 to 15	11 to 15		
Mediterranean California Mixed Evergreen Forest					6 to 10	16 to 20		
Northwestern Great Plains Mixedgrass Prairie	2 to 5	2 to 5						
Rocky Mountain Subalpine-Montane Mesic Meadow	30 <	30 <	30 <	30 <	30 <	30 <		
Ozark-Ouachita Dry-Mesic Oak Forest							6 to 10	6 to 10
Northern Rocky Mountain Montane-Foothill Deciduous Shrubland	6 to 10	11 to 15		11 to 15	11 to 15	6 to 10		
Northern Rocky Mountain Subalpine Deciduous Shrubland	6 to 10	11 to 15		11 to 15		11 to 15		
Southern Rocky Mountain Dry-Mesic Montane Mixed Conifer Forest and Woodland		16 to 20	11 to 15	16 to 20				
Mediterranean California Red Fir Forest				11 to 15	11 to 15	11 to 15		
Inter-Mountain Basins Big Sagebrush Shrubland	11 to 15	11 to 15	11 to 15	11 to 15	11 to 15	11 to 15		
Southeastern North American Temperate Forest Plantation							2 to 5	2 to 5

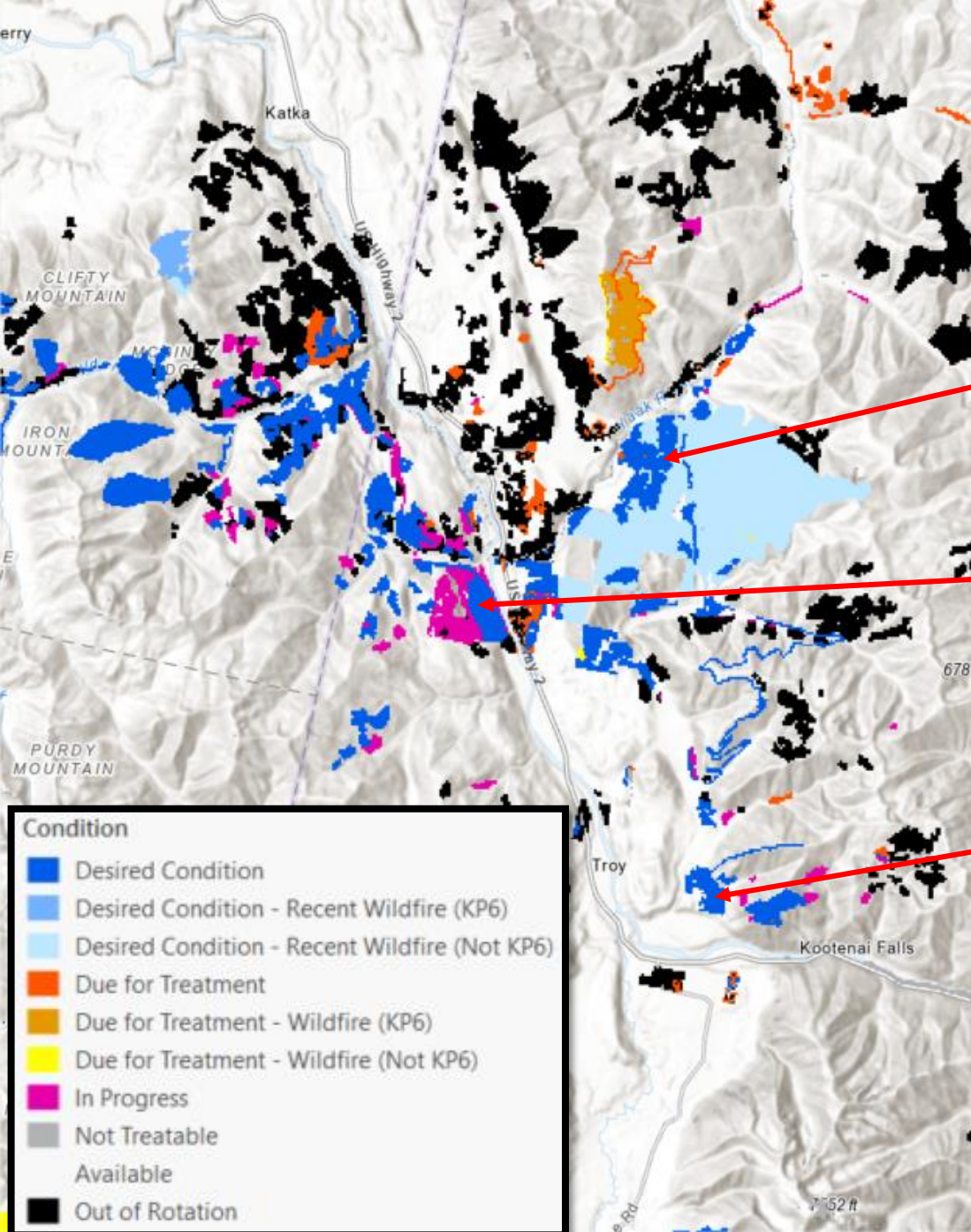


Exploring an example Condition Raster (2025)



Desired Condition

Any non-wildfire FACTS treatments in FP-FUELS-ALL-MIT, including recent Rx fires. We consider these to be in the desired state with respect to risk.



Pile Burn 2024

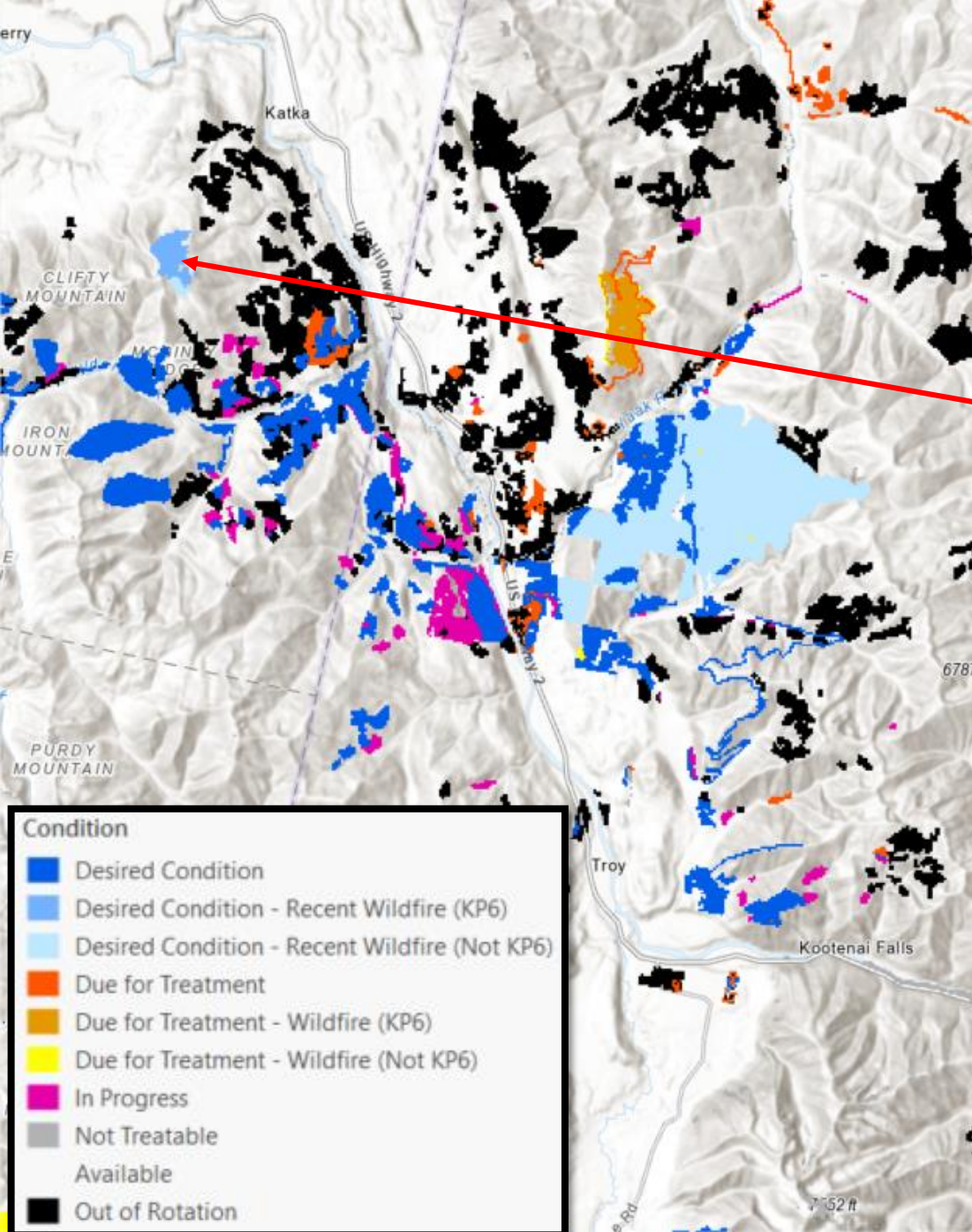
Broadcast Burn 2023

Underburn 2024

All have
PURPOSE_CODE = FTF

Desired Condition – Recent Wildfire (KP6)

Any recent wildfires in FACTS with KP6. These are in the desired state with respect to risk.



Katka Fire 2022

ACTIVITY = "Wildfire - Natural Ignition"

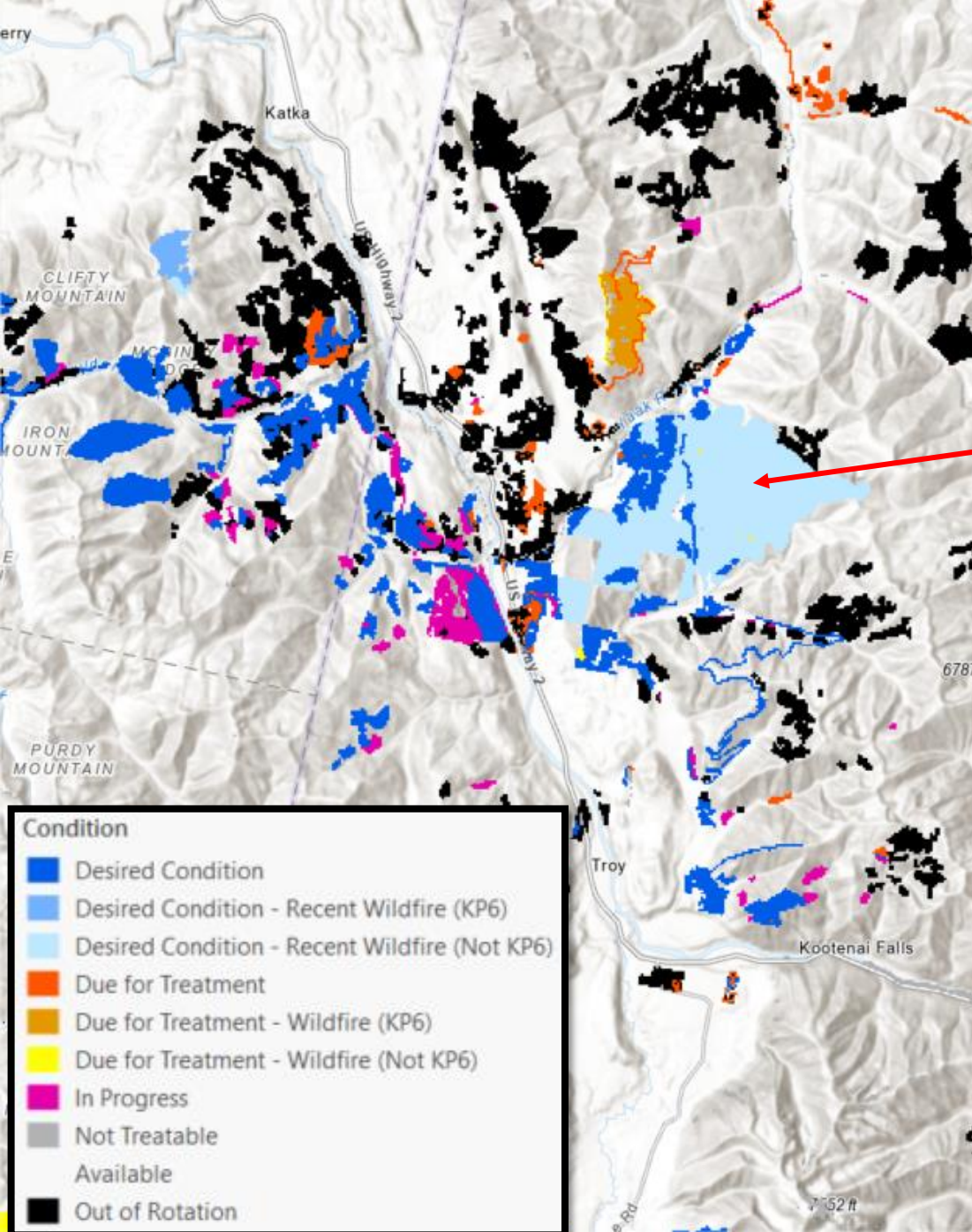
FUELS_KEYPOINT_AREA = "6"

NFPORS_TREATMENT = "Fire Use"

PURPOSE_CODE = FTF

Desired Condition – Recent Wildfire (Not KP6)

Any recent wildfires NOT in KP6. These are in the desired state with respect to risk.



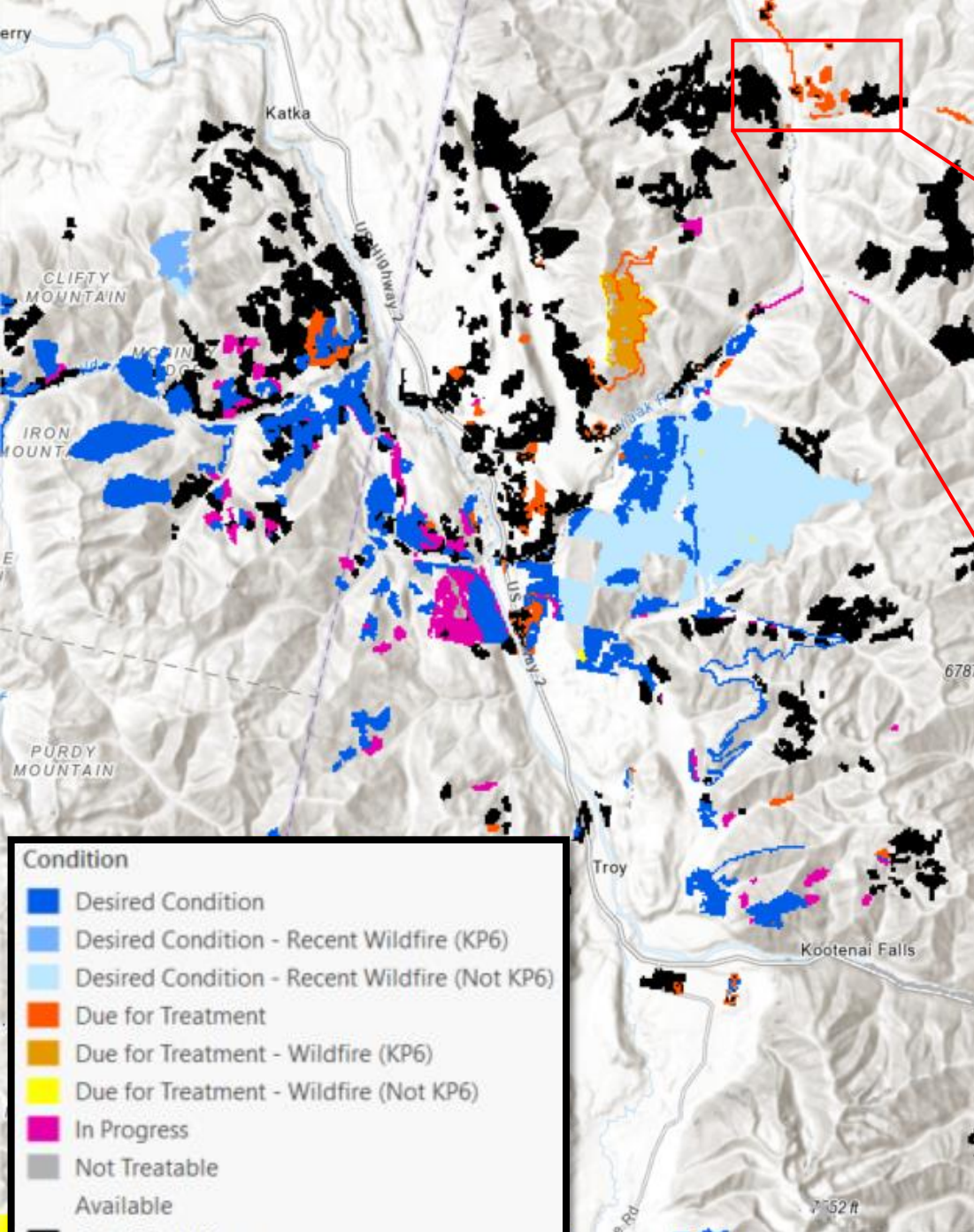
South Yaak Fire 2021

ACTIVITY = "Wildfire - Natural Ignition"

FUELS_KEYPOINT_AREA = <Null>

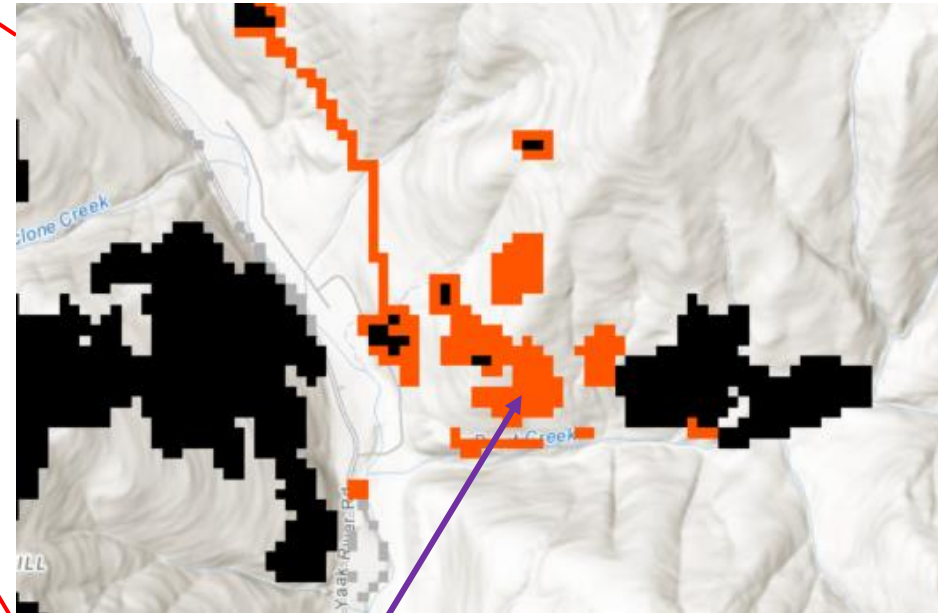
NFPORS_TREATMENT = <Null>

PURPOSE_CODE = <Null>



Due for Treatment

Treated area that has not had a treatment within the defined re-entry period.

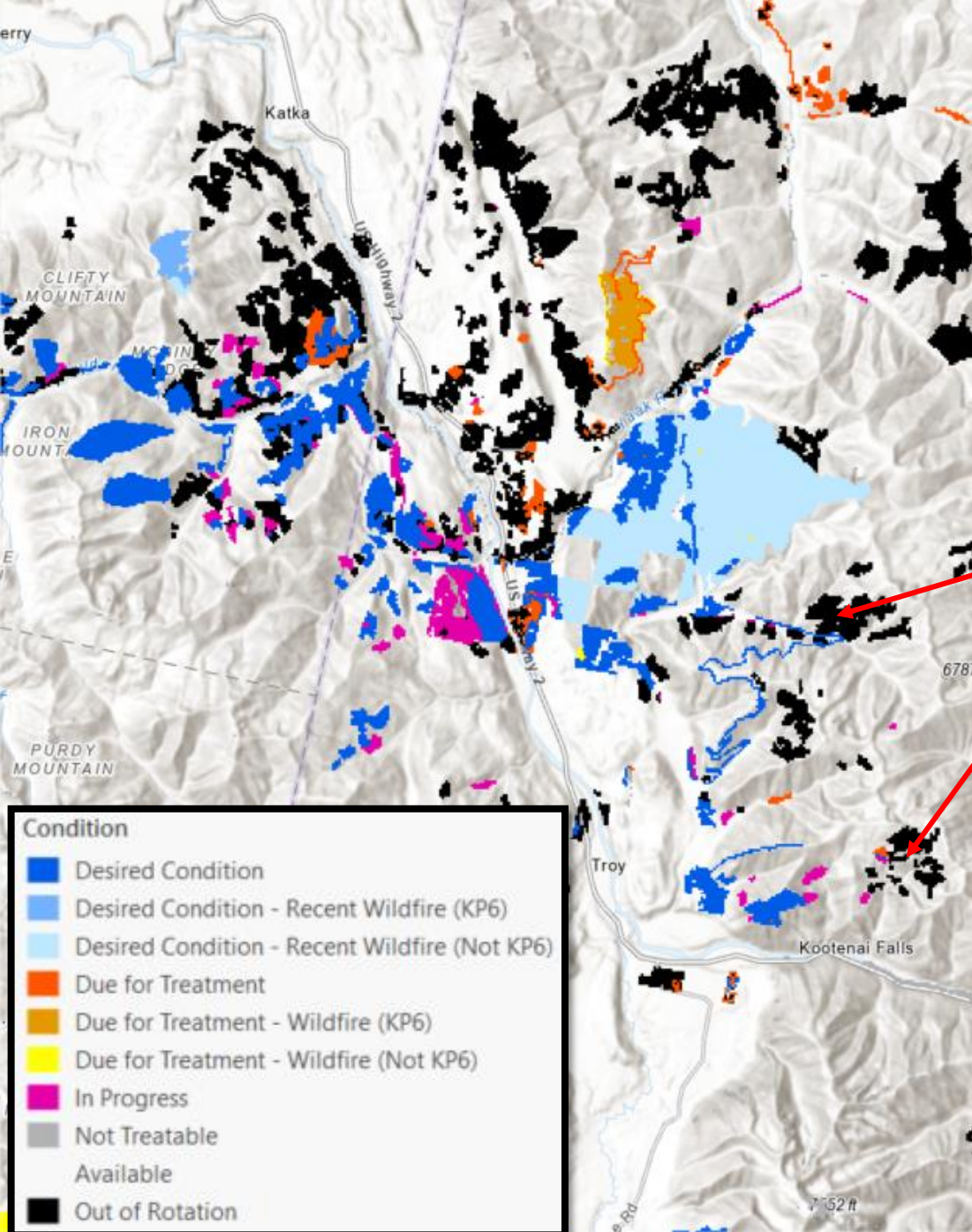


“GRIZZLY 49”

ACTIVITY = “Burning of Piled Material”

FISCAL_YEAR_COMPLETED = 2018

In this EVT, Rotation = 6 - 10 years. Since the treatment is 6 years old, this area is “Due for Treatment”.



Out of Rotation

Any mitigation treatment in the 30-year FACTS record that is older than the upper range of the rotation interval.

Pile burns completed in 1999

In this EVT, Rotation = 6 -10 years. Since the treatments are over 10 years, these areas are considered “Out of Rotation”.

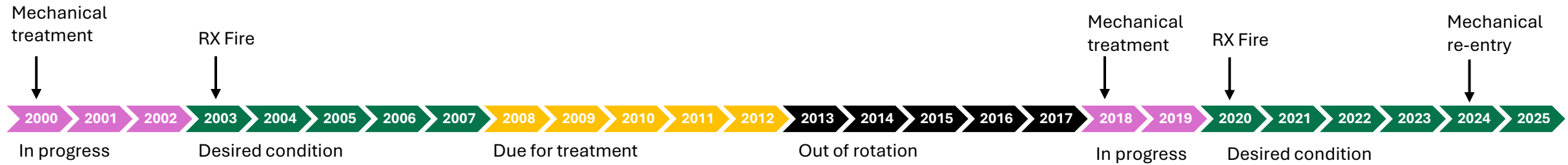
Annual Accounting



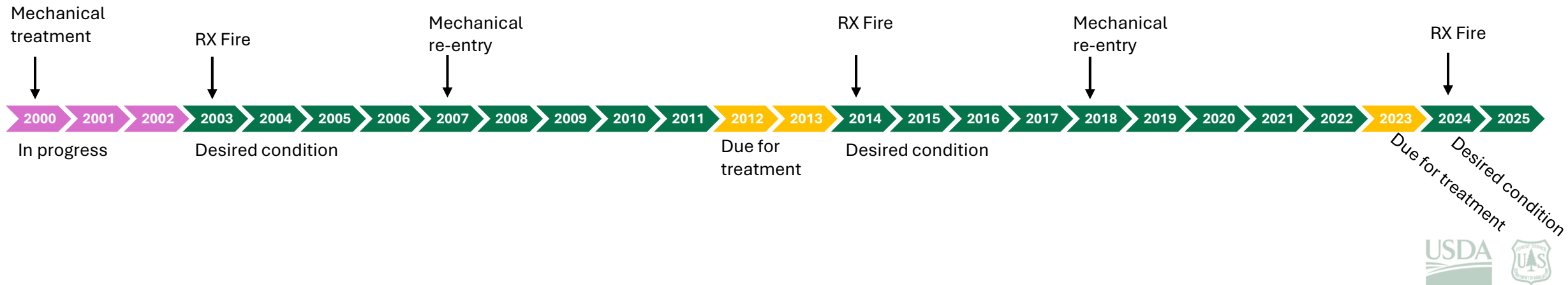
How do we account for annual changes in treatment condition?

Annual Treatment Accounting

6 to 10-year rotation

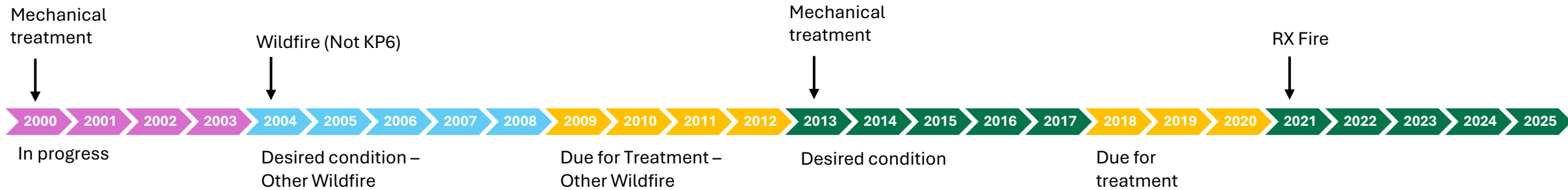


6 to 10-year rotation

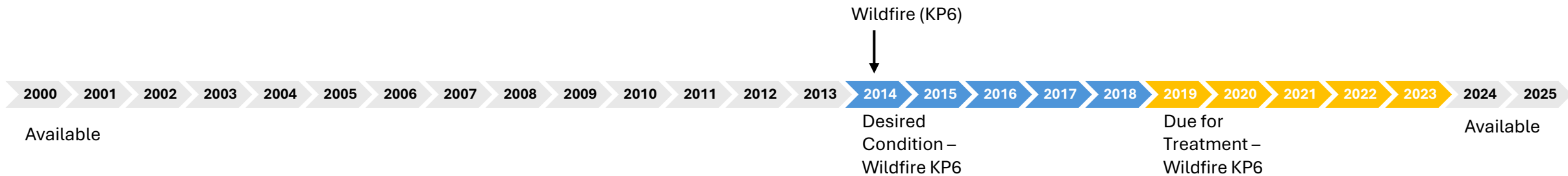


Annual Treatment Accounting

6 to 10-year rotation

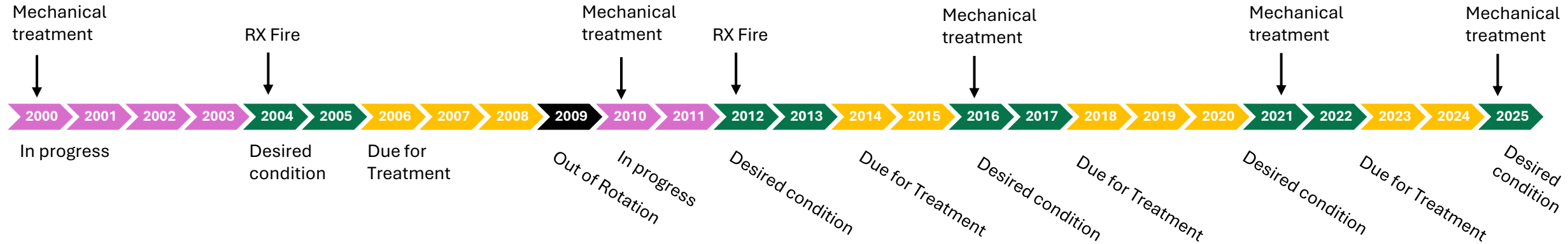


6 to 10-year rotation

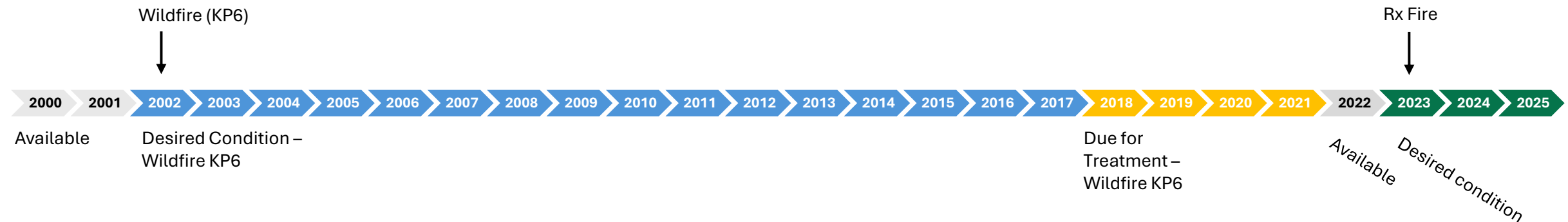


Annual Treatment Accounting

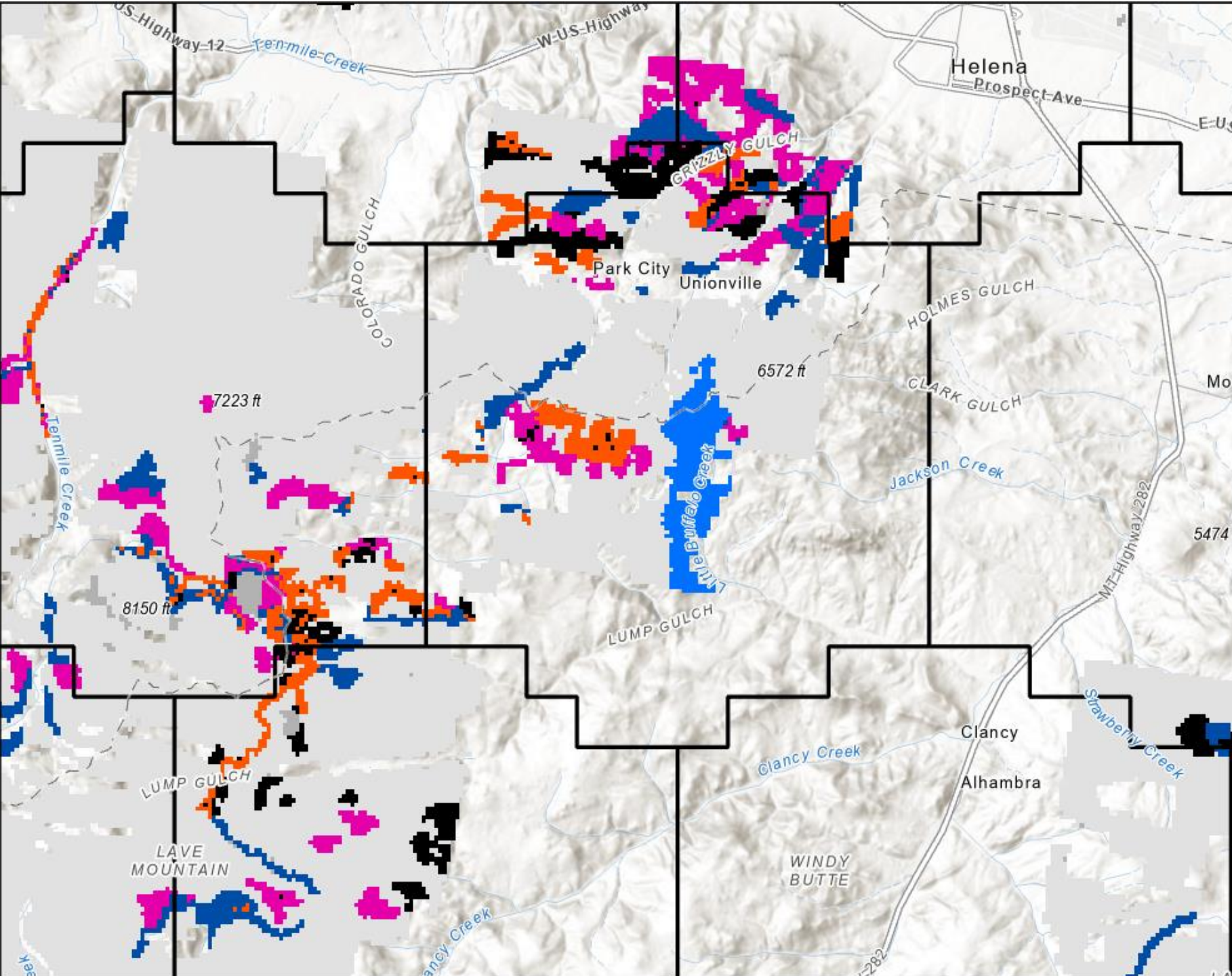
2 to 5-year rotation



16 to 20-year rotation



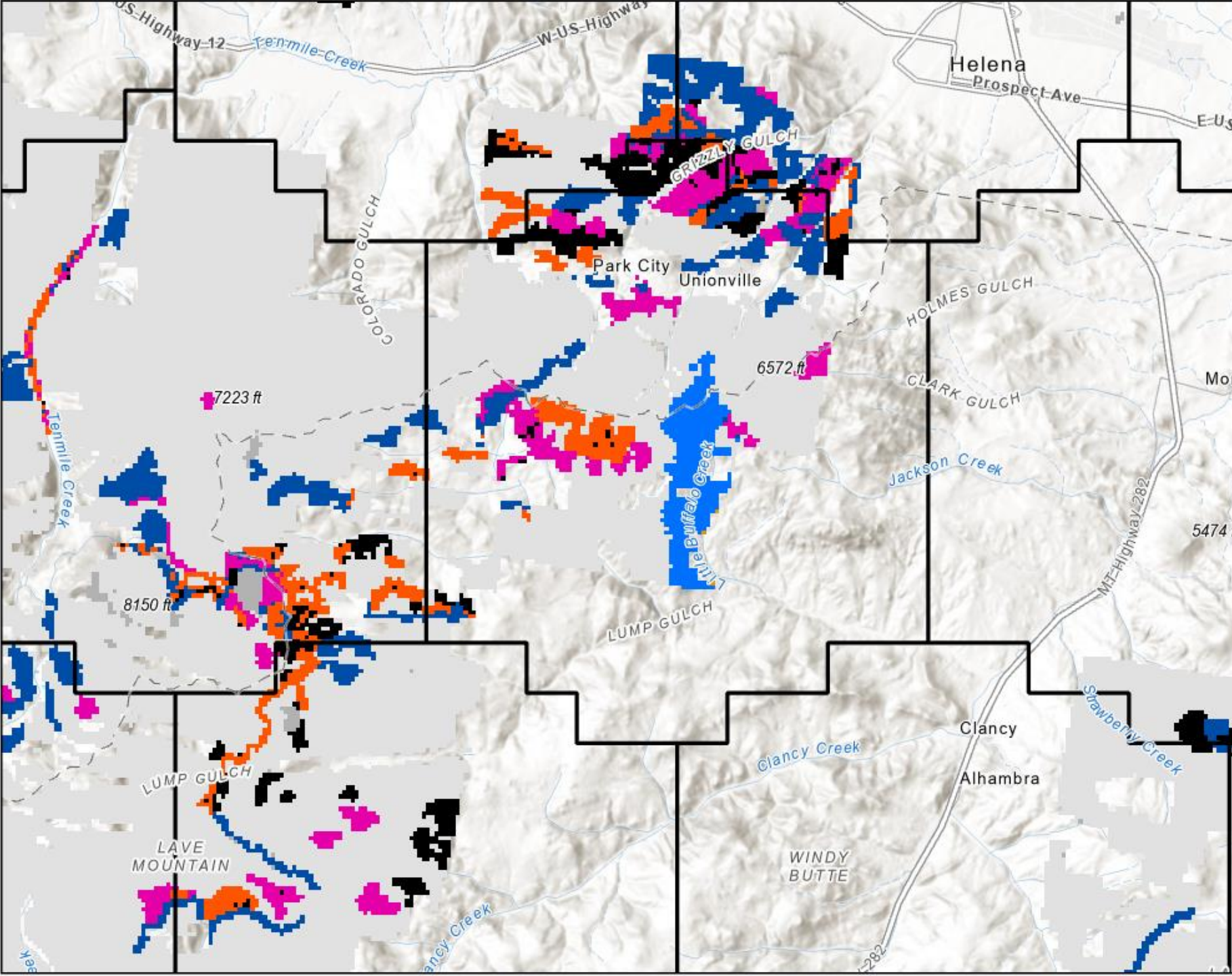
2021



Condition

- Available
- Desired Condition
- Desired Condition - Recent Wildfire (KP6)
- Desired Condition - Recent Wildfire (Not KP6)
- Due for Treatment
- Due for Treatment - Wildfire (KP6)
- Due for Treatment - Wildfire (Not KP6)
- In Progress
- Not Treatable
- Out of Rotation

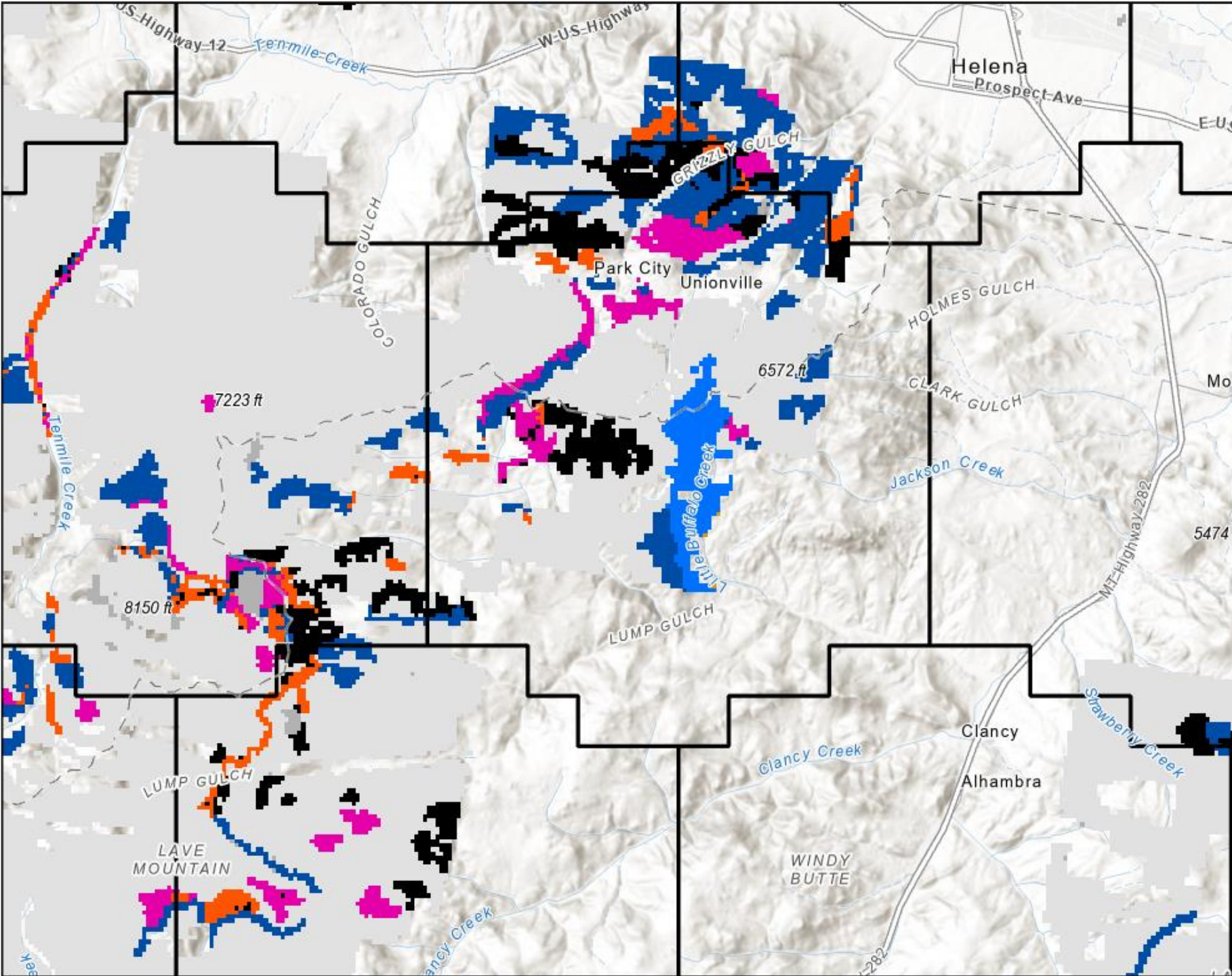
2022



Condition

- Available
- Desired Condition
- Desired Condition - Recent Wildfire (KP6)
- Desired Condition - Recent Wildfire (Not KP6)
- Due for Treatment
- Due for Treatment - Wildfire (KP6)
- Due for Treatment - Wildfire (Not KP6)
- In Progress
- Not Treatable
- Out of Rotation

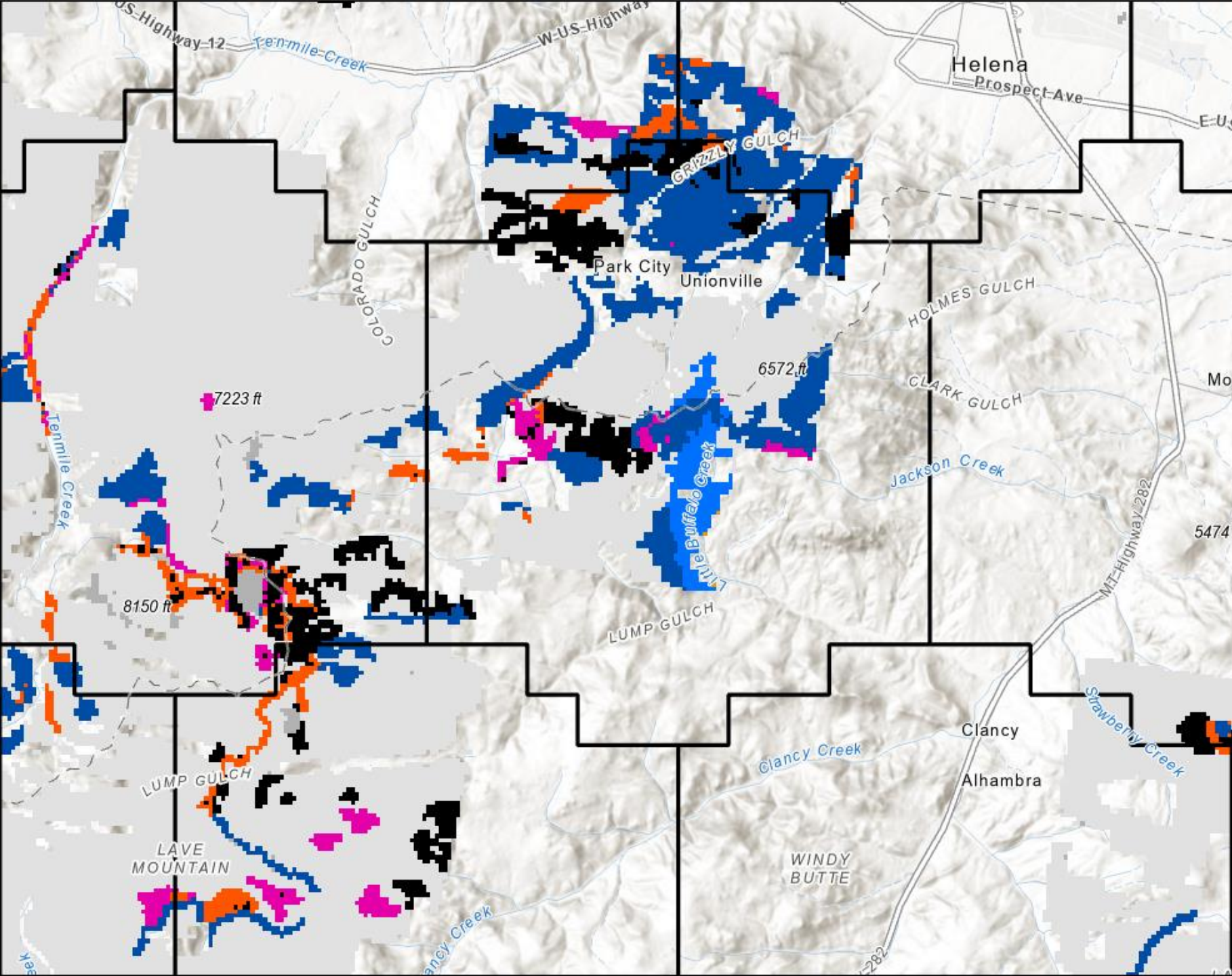
2023



Condition

- Available
- Desired Condition
- Desired Condition - Recent Wildfire (KP6)
- Desired Condition - Recent Wildfire (Not KP6)
- Due for Treatment
- Due for Treatment - Wildfire (KP6)
- Due for Treatment - Wildfire (Not KP6)
- In Progress
- Not Treatable
- Out of Rotation

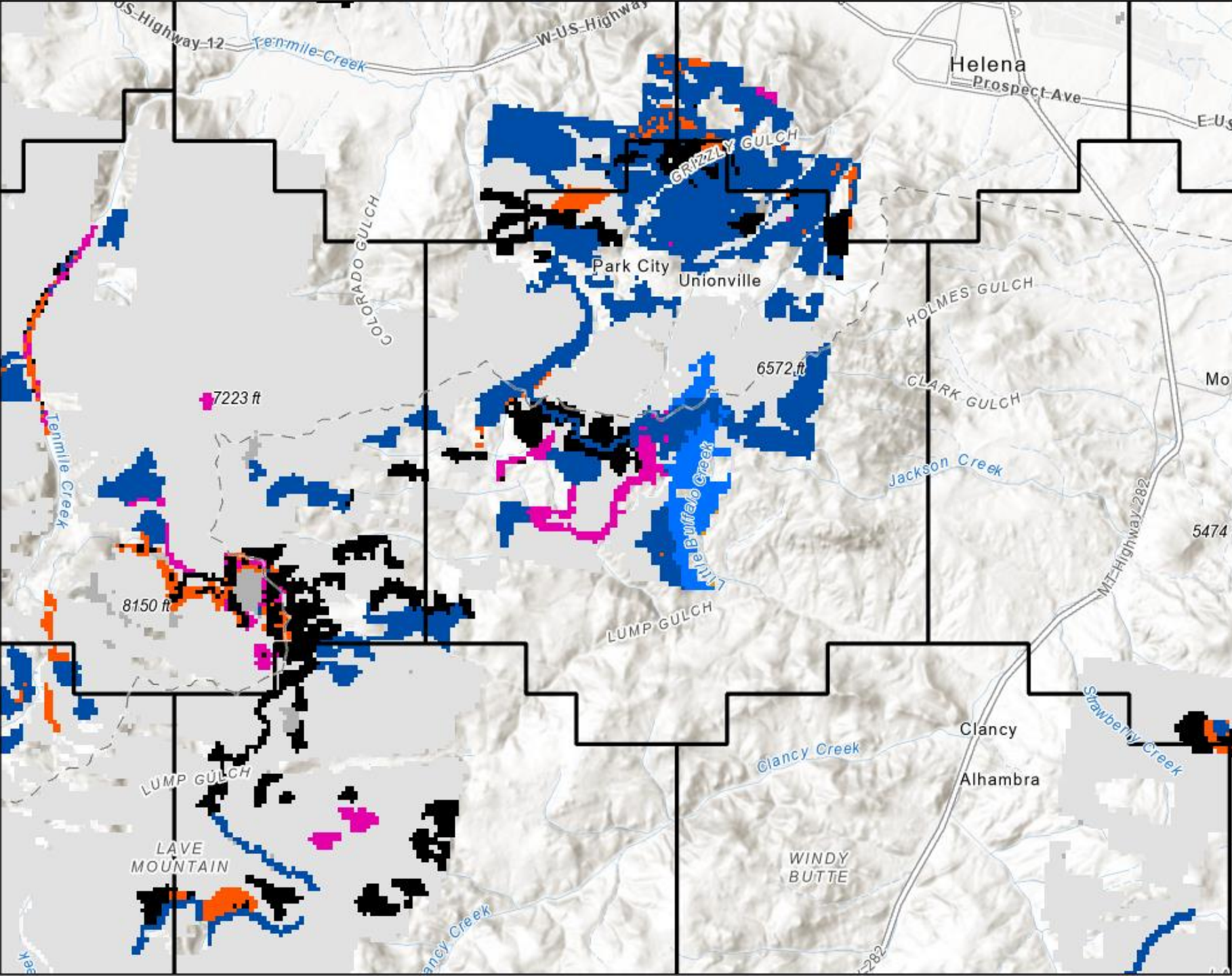
2024



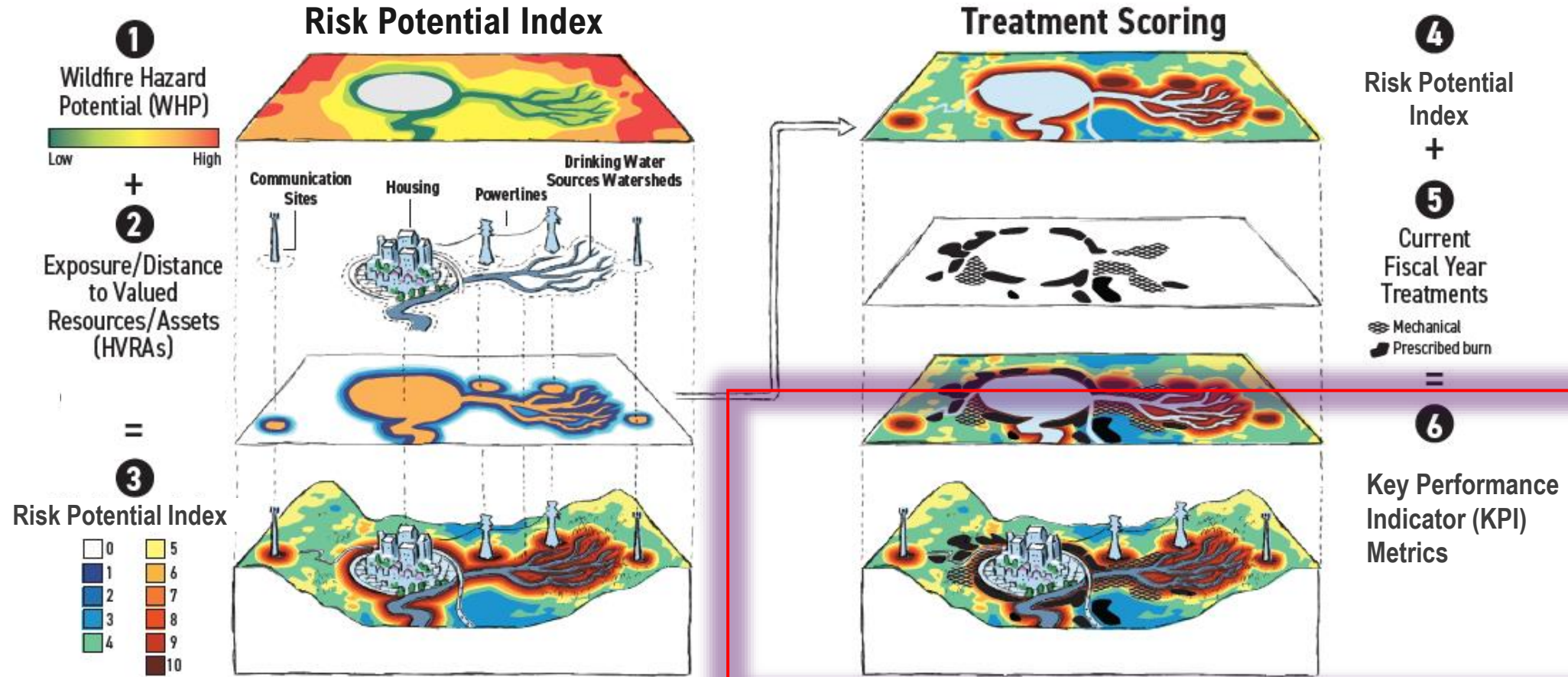
Condition

- Available
- Desired Condition
- Desired Condition - Recent Wildfire (KP6)
- Desired Condition - Recent Wildfire (Not KP6)
- Due for Treatment
- Due for Treatment - Wildfire (KP6)
- Due for Treatment - Wildfire (Not KP6)
- In Progress
- Not Treatable
- Out of Rotation

2025



WRR-KPI Summary Metrics



KPI Metric 1: DLRC Score

% of RPI in Desired Low Risk Condition (DLRC)

$$\% \text{ RPI in DLRC} = \frac{\text{RPI in DLRCT} + \text{RPI in DLRCWF}}{\text{Total RPI}}$$

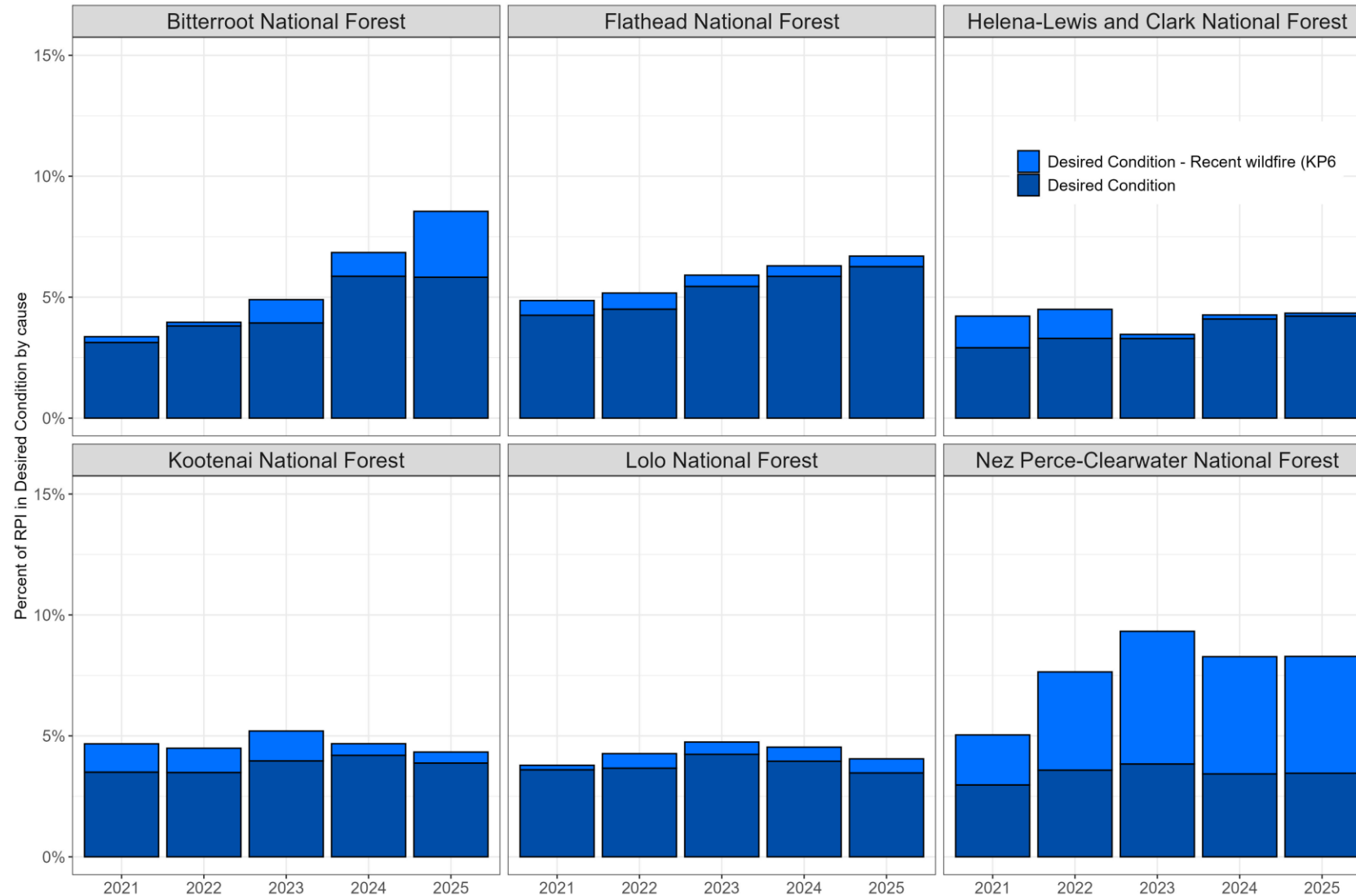
$DLRC_T$ = in DLRC due to treatment

$DLRC_{WF}$ = in DLRC due to KP6 Wildfire

- This metric is **cumulative across multiple years** and can be used as a progress indicator.
- This metric considers the **condition of treatments from previous years**.
- Adding treatments that mitigate risk in high RPI and maintaining previous investments will increase this metric.

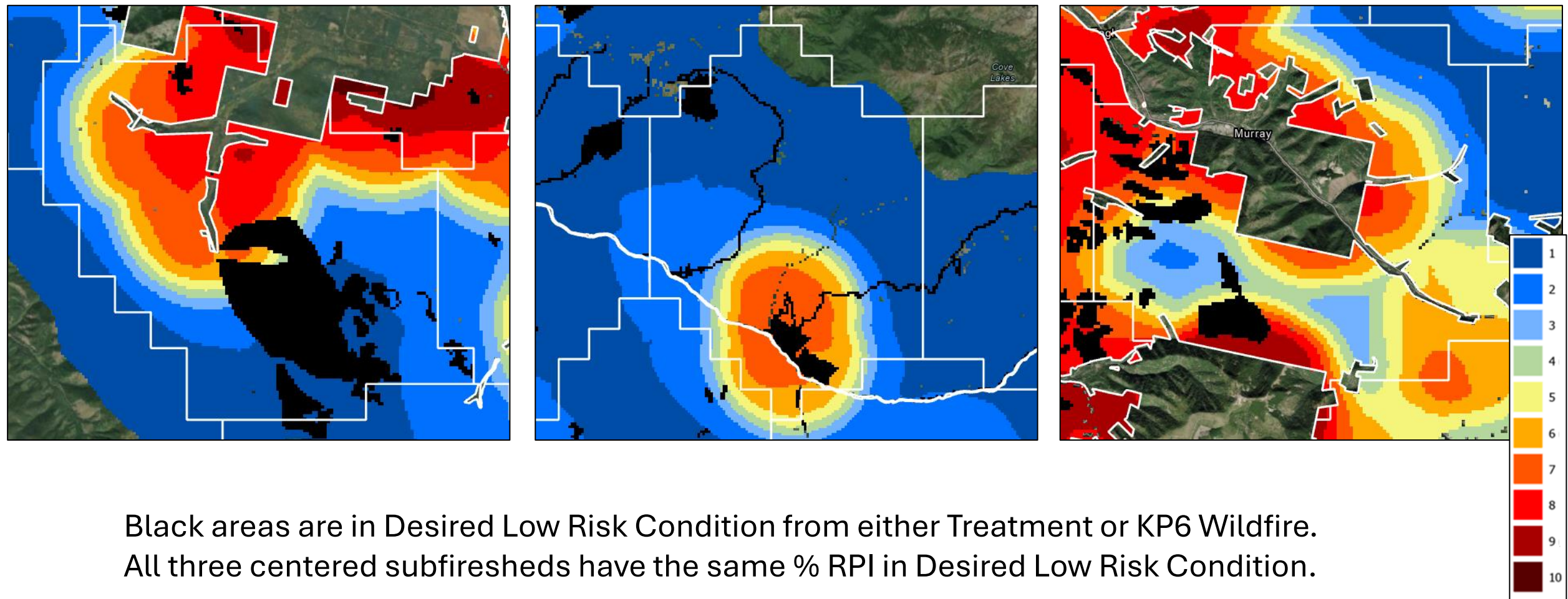
KPI Metric 1: DLRC Score

% of RPI in Desired Low Risk Condition (DLRC)



KPI Metric 1: % of RPI in Desired Low Risk Condition

10 %



KPI Metric 2: Risk Treatment Optimization Score

Average RPI Treated

$$\text{Average RPI treated} = \frac{\text{sum of treated RPI}}{\text{count of treated pixels}}$$

- This metric is **cumulative within a year** can be used as a progress and targeting indicator.
- This metric does not consider treatments from previous years.
- We include **intentional treatments** (non-wildfire) that are in **Desired Low Risk Condition** or **In Progress** for this metric.
- Performing treatments in higher RPI areas will increase this metric.

KPI Metric 2: Risk Treatment Optimization Score

Average RPI Treated



KPI Metric 2: Risk Treatment Optimization Score

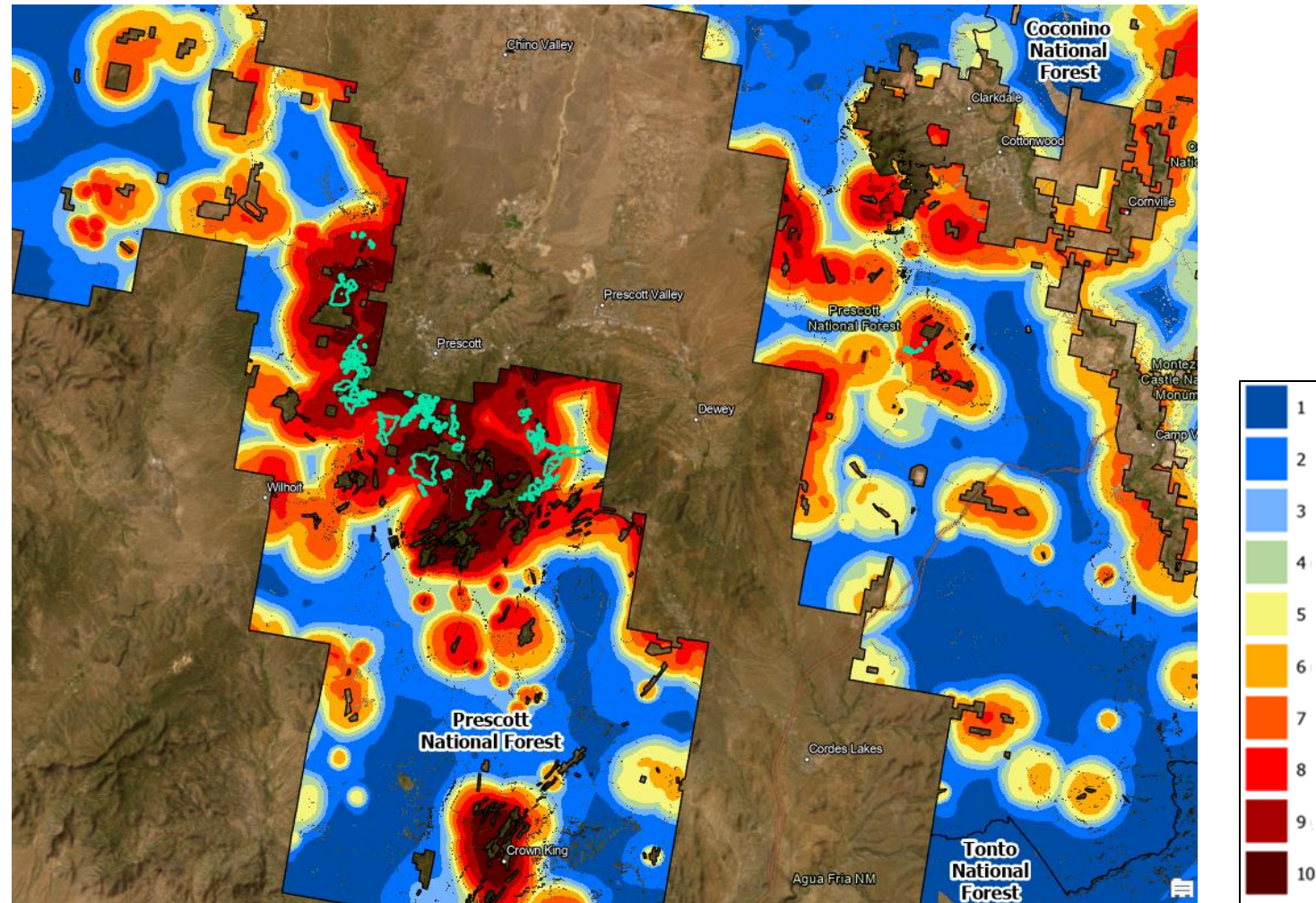
Average RPI Treated

	Forest RPI Average	Acres					RPI Average				
		2021	2022	2023	2024	2025	2021	2022	2023	2024	2025
Apache-Sitgreaves National Forests	2.9	 22,513	 49,520	 43,570	 70,403	 33,802	4.0	2.9	2.5	2.8	3.4
Carson National Forest	2.8	 16,199	 11,315	 6,193	 19,903	 13,919	2.7	2.1	3.6	2.9	3.2
Cibola National Forest	2.1	 7,862	 6,005	 14,309	 15,526	 13,436	3.7	4.7	2.9	3.4	4.4
Coconino National Forest	4.2	 10,660	 8,627	 17,588	 44,244	 32,101	6.2	7.5	5.6	6.9	6.4
Coronado National Forest	2.5	 14,221	 17,778	 8,747	 8,899	 6,891	2.4	1.0	3.3	4.5	6.5
Gila National Forest	2.2	 23,236	 3,887	 20,672	 16,072	 3,387	2.1	5.3	2.6	3.1	4.2
Kaibab National Forest	2.1	 10,460	 30,680	 49,935	 55,989	 32,713	5.1	3.7	3.6	3.4	2.8
Lincoln National Forest	4.2	 1,261	 1,145	 1,413	 2,176	 2,540	6.8	8.1	7.3	7.8	7.4
Prescott National Forest	2.8	 7,438	 13,500	 15,358	 25,324	 23,142	8.6	5.5	5.7	4.0	4.3
Santa Fe National Forest	3.7	 10,748	 10,042	 4,670	 6,481	 3,791	4.0	3.4	3.5	3.5	2.5
Tonto National Forest	3.3	 7,748	 31,661	 14,483	 21,681	 17,890	6.5	6.0	6.9	7.8	7.5

KPI Metric 2: Risk Treatment Optimization Score

Average RPI Treated

Prescott National Forest in 2021



KPI Metric 3: Effective Acres or Weighted Acres

Total RPI Treated

Total RPI Treated = sum of Treated RPI

- This metric is **cumulative within a year** can be used as a progress and targeting indicator.
- This metric does not consider treatments from previous years.
- We include **intentional treatments** (non-wildfire) that are in **Desired Low Risk Condition** or **In Progress** for this metric.
- Performing a greater amount of treatment in high RPI areas will increase this metric.

KPI Metric 3: Effective Acres or Weighted Acres

Total RPI Treated



KPI Metric 3: Effective Acres or Weighted Acres

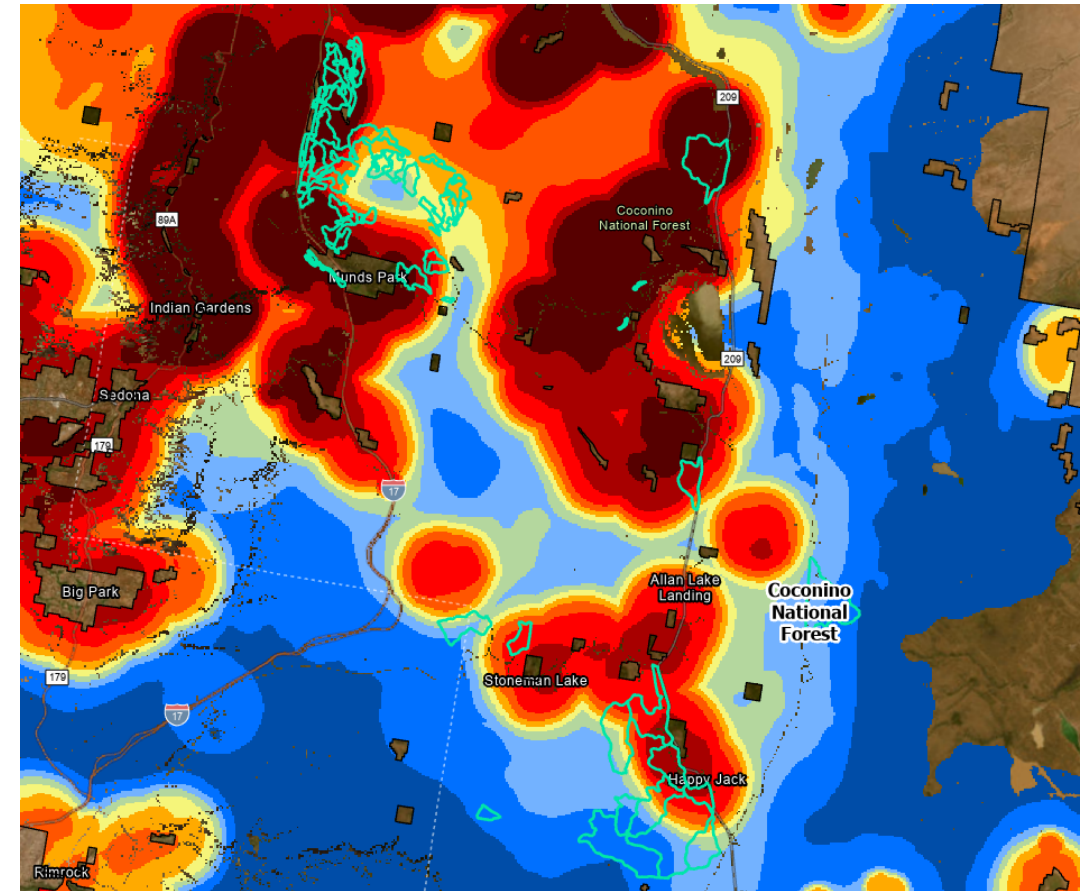
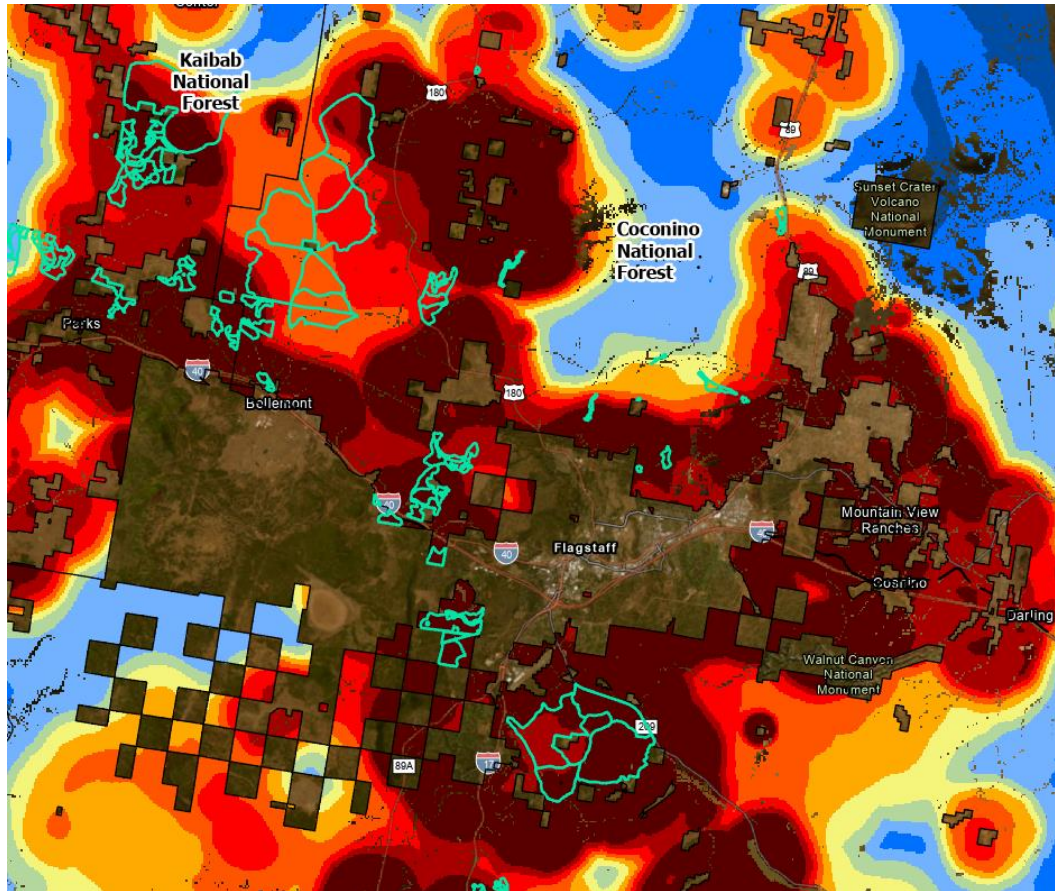
Total RPI Treated

	Acres					RPI Average					RPI Sum				
	2021	2022	2023	2024	2025	2021	2022	2023	2024	2025	2021	2022	2023	2024	2025
Apache-Sitgreaves National Forests	22,513	49,520	43,570	70,403	33,802	4.0	2.9	2.5	2.8	3.4	43,759	71,593	49,406	97,380	56,145
Carson National Forest	16,199	11,315	6,193	19,903	13,919	2.7	2.1	3.6	2.9	3.2	21,244	11,849	11,173	28,475	21,973
Cibola National Forest	7,862	6,005	14,309	15,526	13,436	3.7	4.7	2.9	3.4	4.4	14,051	13,589	18,172	25,867	29,004
Coconino National Forest	10,660	8,627	17,588	44,244	32,101	6.2	7.5	5.6	6.9	6.4	32,148	31,129	47,492	149,301	99,648
Coronado National Forest	14,221	17,778	8,747	8,899	6,891	2.4	1.0	3.3	4.5	6.5	15,216	8,520	13,426	19,207	22,134
Gila National Forest	23,236	3,887	20,672	16,072	3,387	2.1	5.3	2.6	3.1	4.2	23,278	9,643	26,788	23,928	6,627
Kaibab National Forest	10,460	30,680	49,935	55,989	32,713	5.1	3.7	3.6	3.4	2.8	24,577	54,647	86,492	94,006	44,103
Lincoln National Forest	1,261	1,145	1,413	2,176	2,540	6.8	8.1	7.3	7.8	7.4	4,231	4,611	5,094	8,260	9,361
Prescott National Forest	7,438	13,500	15,358	25,324	23,142	8.6	5.5	5.7	4.0	4.3	30,746	36,602	42,732	48,919	48,438
Santa Fe National Forest	10,748	10,042	4,670	6,481	3,791	4.0	3.4	3.5	3.5	2.5	21,308	16,804	8,090	11,405	4,807
Tonto National Forest	7,748	31,661	14,483	21,681	17,890	6.5	6.0	6.9	7.8	7.5	24,326	93,069	47,582	81,761	65,028

KPI Metric 3: Effective Acres or Weighted Acres

Total RPI Treated

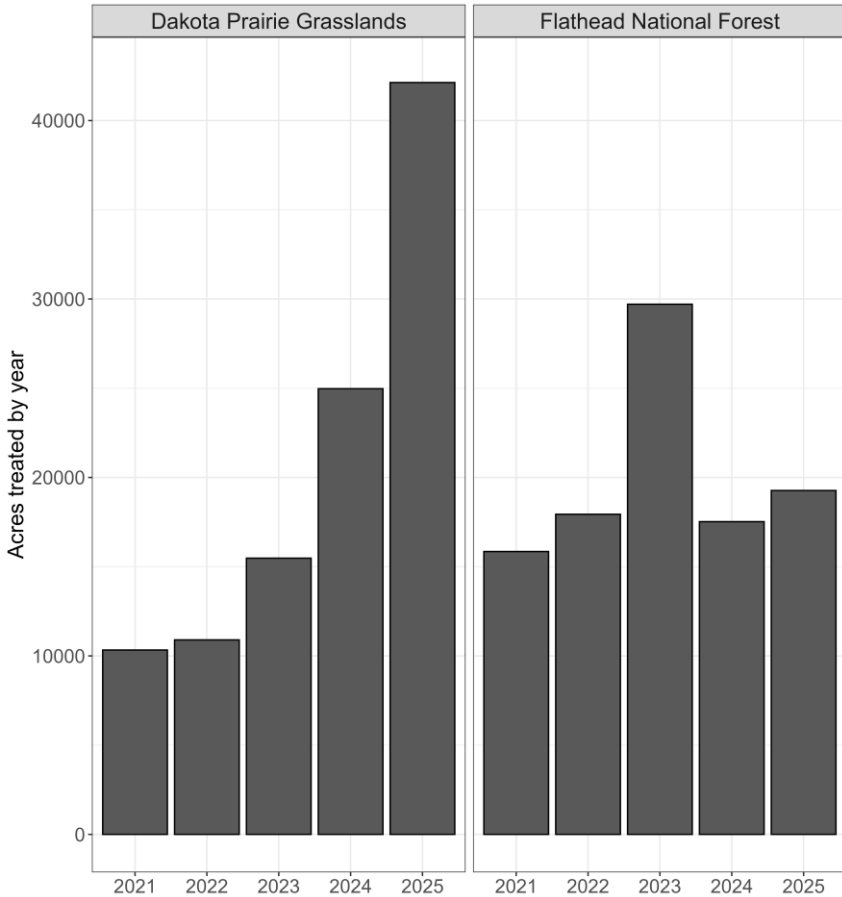
Coconino National Forest - 2024



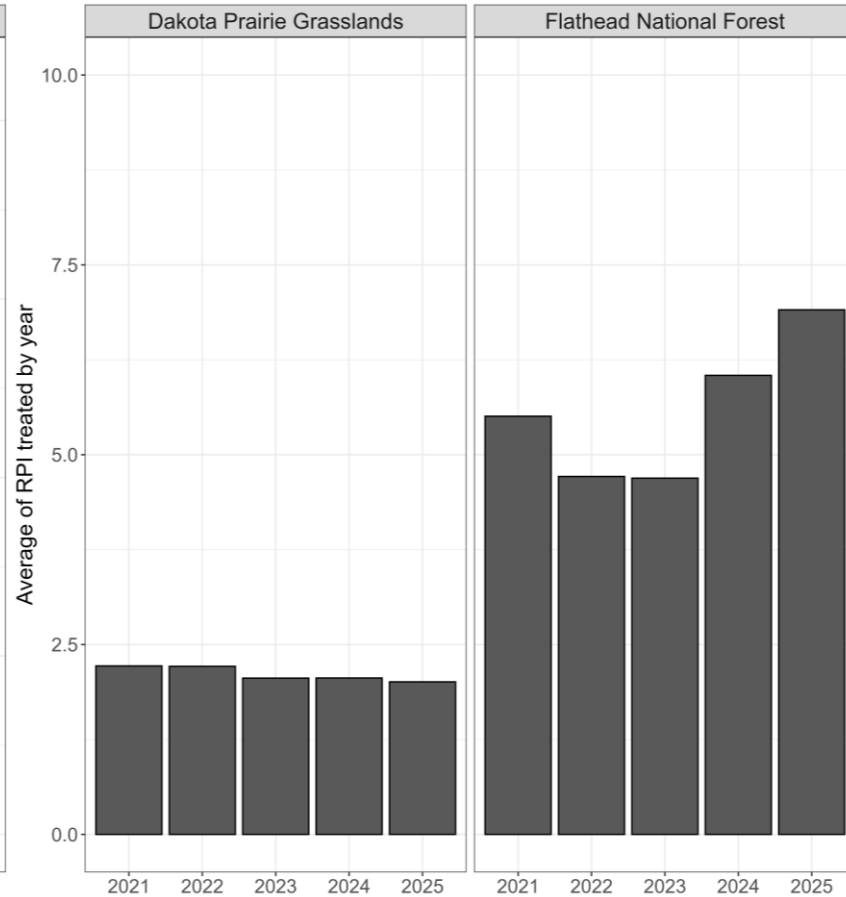
KPI Metrics :

Putting it all together

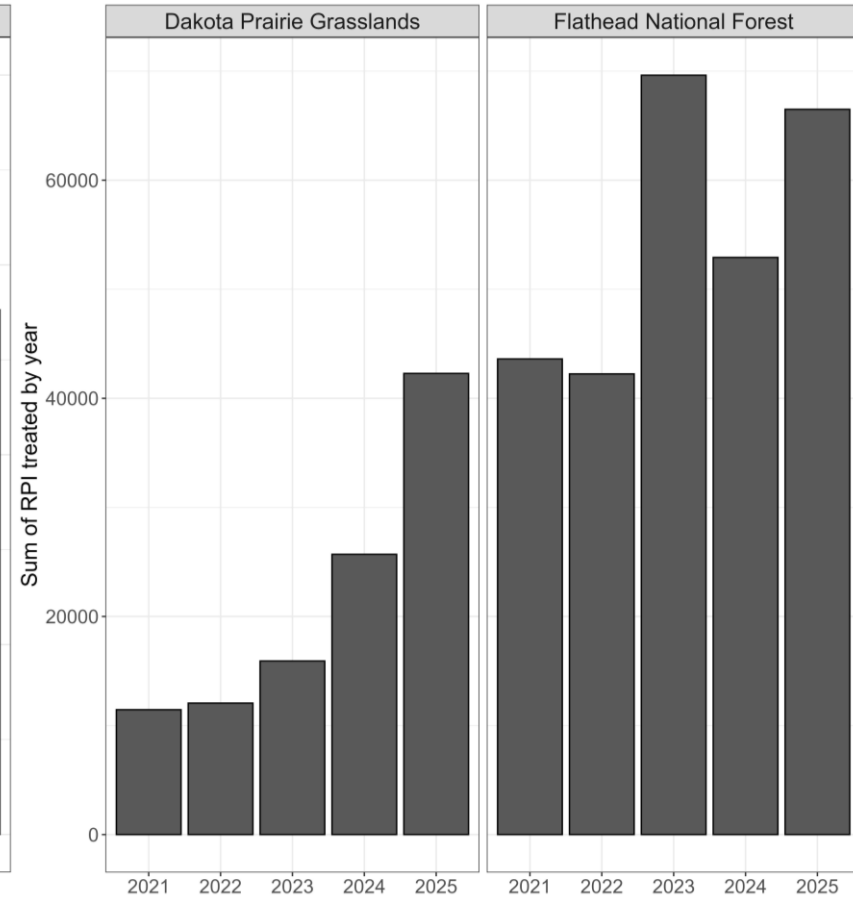
Acres Treated



Average RPI Treated



Total RPI Treated



A group of approximately 15 firefighters in yellow shirts and helmets are gathered in a clearing within a forest. They are standing on a dirt path, and their shadows are cast long on the ground. In the background, there is a large, light-colored structure, possibly a tent or a piece of equipment. The scene is framed by large, dark tree trunks in the foreground, and the sun is shining brightly through the trees, creating a lens flare effect.

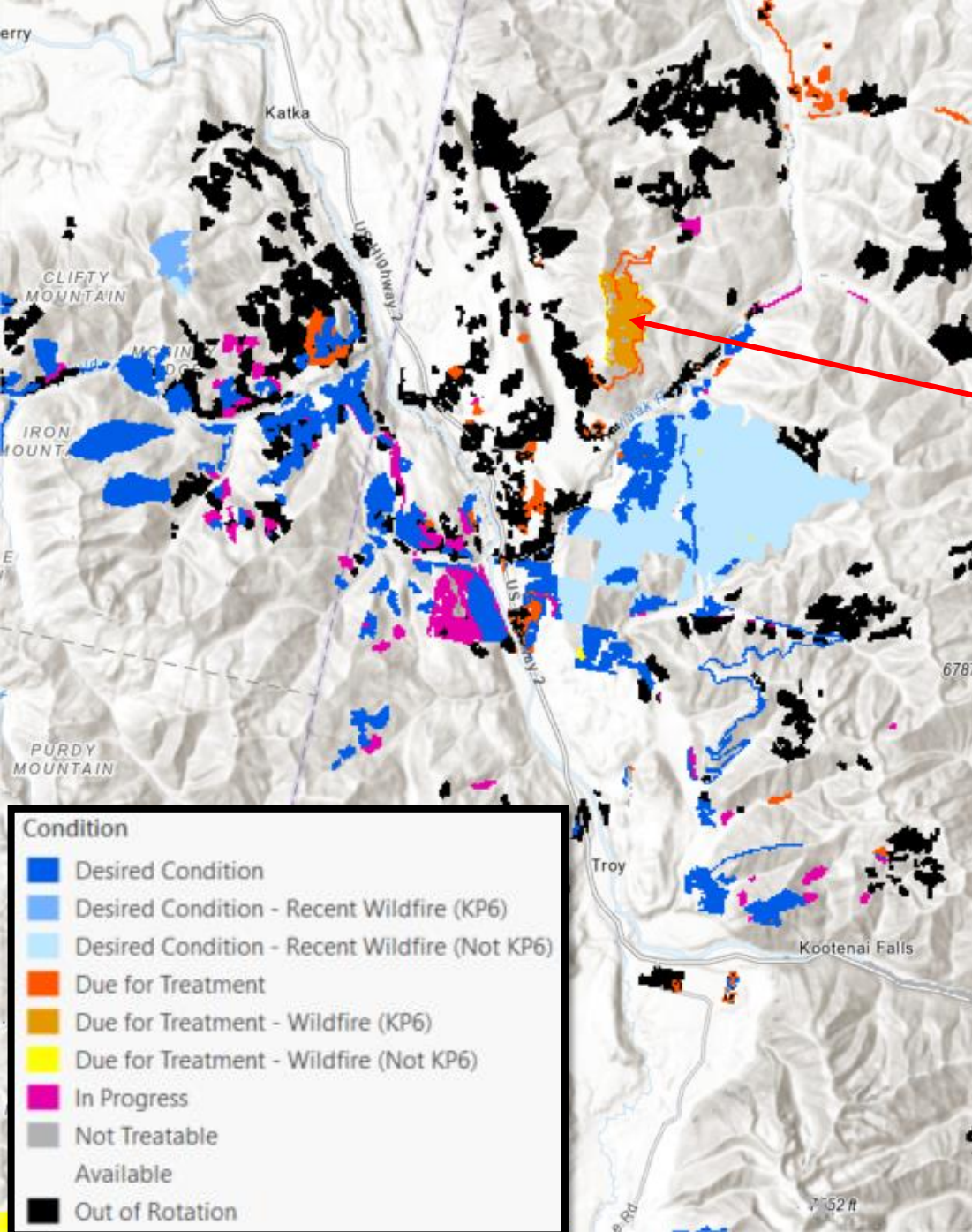
Questions?

An aerial photograph of a forest landscape. A road or path runs vertically through the center-right of the image. The forest is composed of various types of trees, some with green foliage and others with brown, suggesting a mix of species or seasonal changes. A large, solid green rectangular box is superimposed over the center-left portion of the image, containing the word "Extras" in white text.

Extras

Extras – More Examples of Treatment Condition

The following slides provide more specific examples of the different categories in the treatment condition raster



Due for Treatment – Wildfire (KP6)

KP6 Wildfire that has not had a treatment within the defined re-entry period.

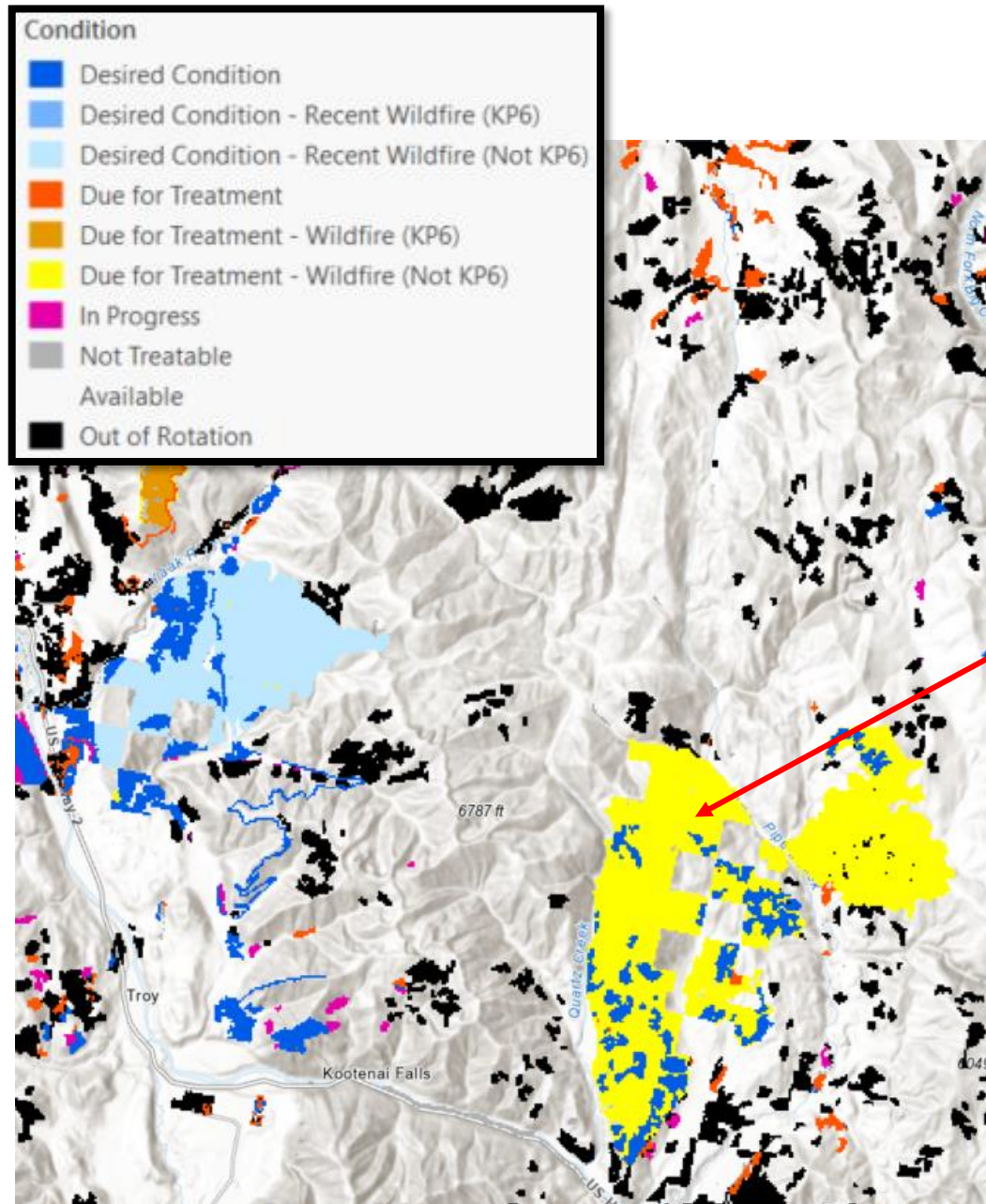
Teepee Fire 2015

ACTIVITY = “Wildfire - Natural Ignition”

FUELS_KEYPOINT_AREA = 6

NFPORS_TREATMENT = “Fire Use”

In this EVT, Rotation = 6 - 10 years. Since the treatment is a 10-year-old KP6 wildfire, this area is “Due for Treatment – Wildfire (KP6)”.



Due for Treatment – Wildfire (Not KP6)

Wildfire NOT in KP6 that has not had a treatment within the defined re-entry period.

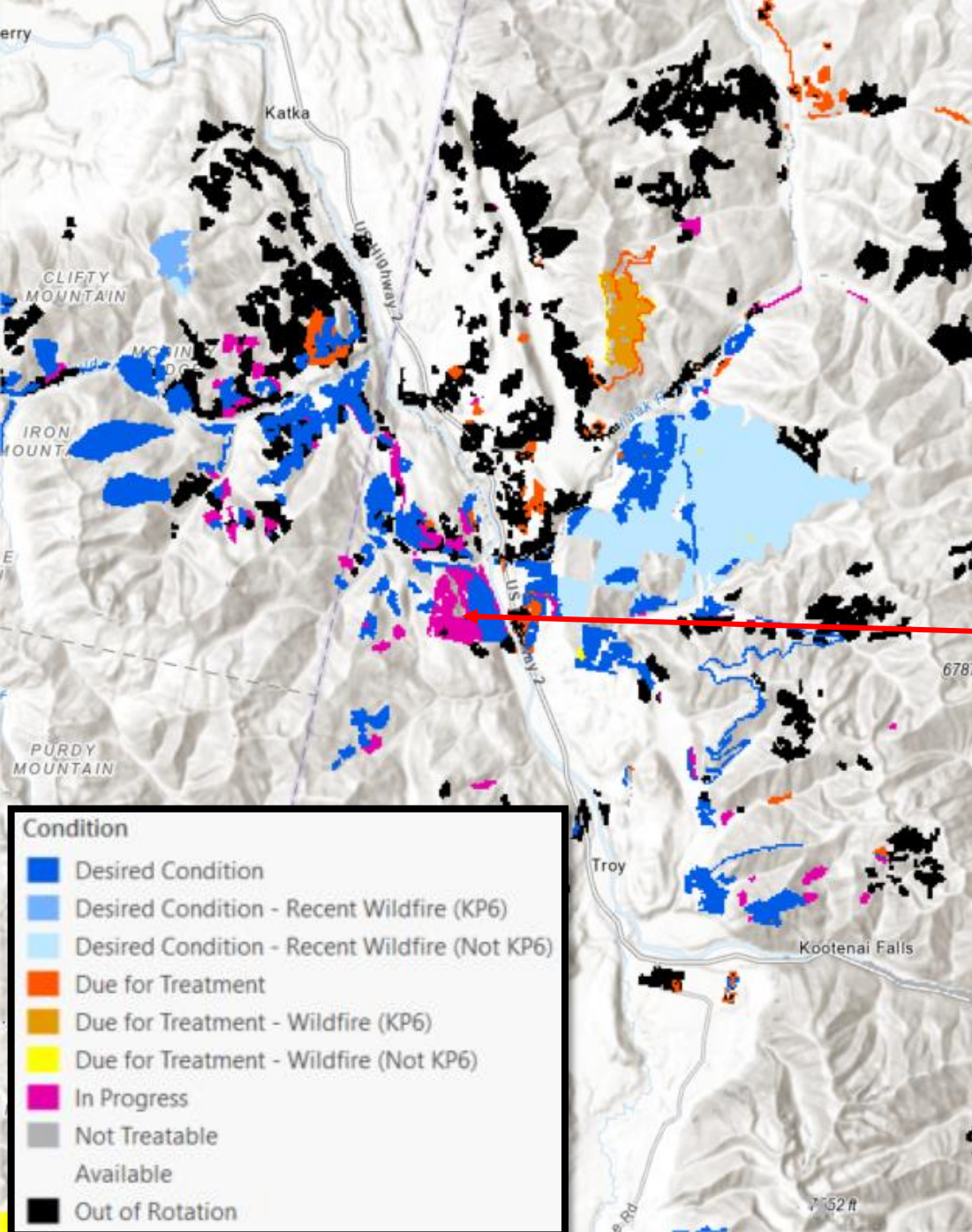
West Fork Fire 2017

ACTIVITY = “Wildfire - Natural Ignition”

FUELS_KEYPOINT_AREA = <Null>

NFPORS_TREATMENT = <Null>

In this EVT, Rotation = 6 - 10 years. Since the treatment is an 8-year-old non-KP6 wildfire, this area is “Due for Treatment – Wildfire (Not KP6)”.

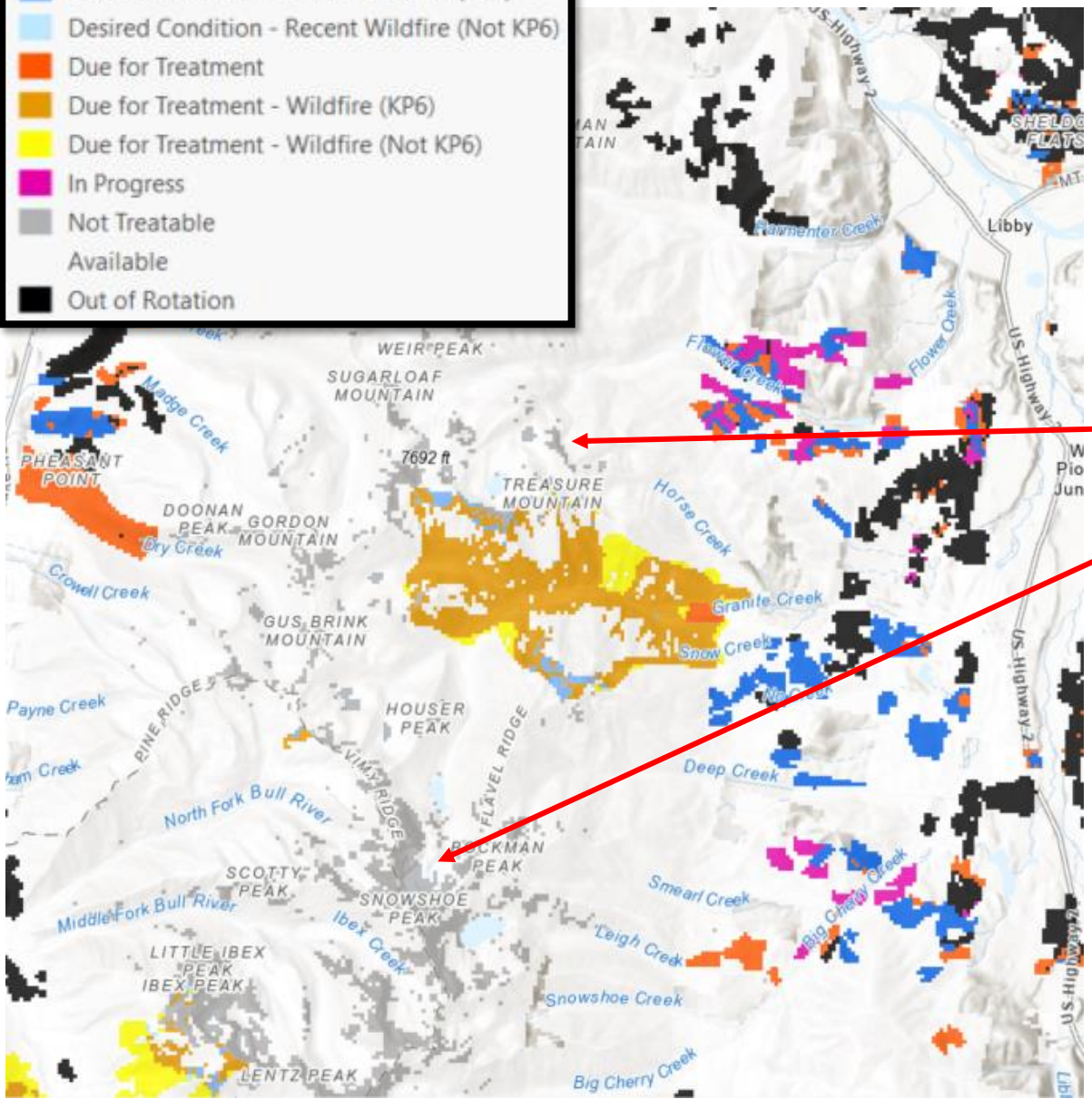


In Progress

Any FACTS treatments in FP-FUELS-ALL, but not marked as mitigated. Has had a treatment within the number of years specified within the upper bound of the rotation period (e.g. in a 6-10 year Rotation Interval, any treatment within the past 10 years will be In Progress).

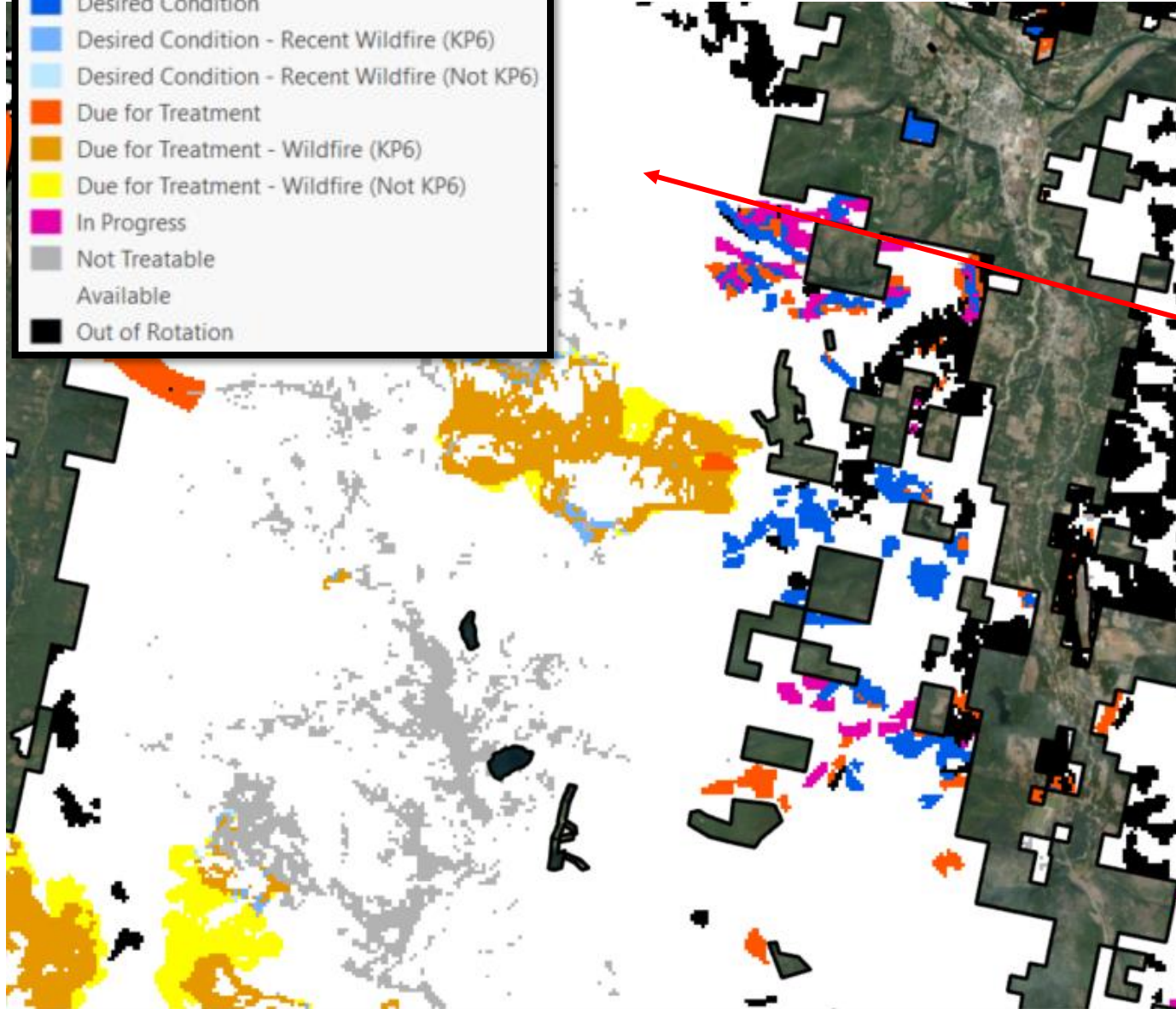
“Starry Goat L 206” Treatment 2024

ACTIVITY = “Rearrangement of Fuels”,
NFPORS_TREATMENT = “Lop and Scatter”
FUELS_KEYPOINT_AREA = 3
PURPOSE_CODE = FTI



Not Treatable

Existing Vegetation Types considered “not treatable” by the region (e.g. Sparse Vegetation).



Available

Forest Service land that has not yet been treated but is available to be treated. (All the white area).

Forecasting max DLRC treatable / maintainable

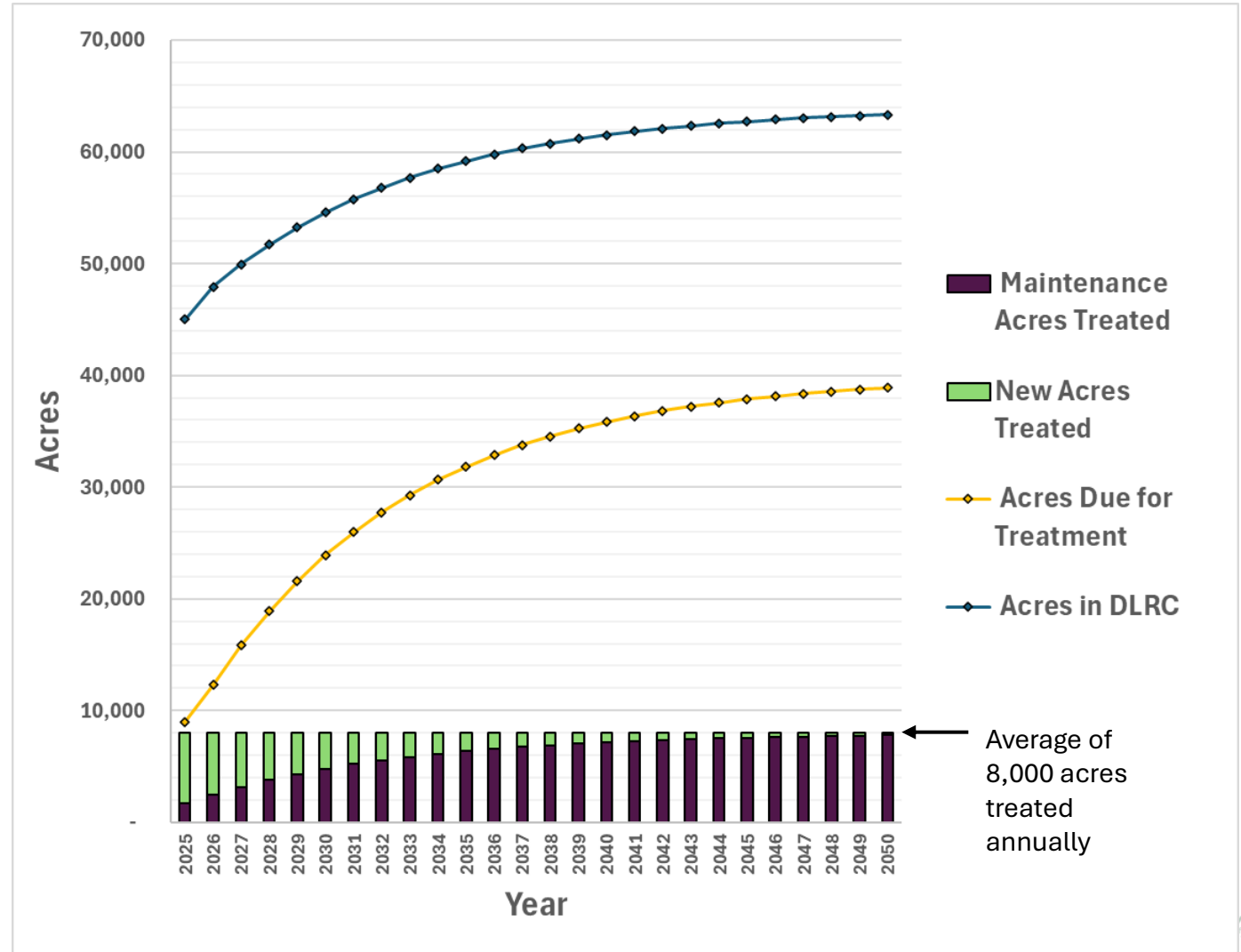
The following slides walk through the logic of using treatment condition and rotation information to forecast the amount of area that can be maintained in DLRC under different levels of annual treatment effort

How many acres can we maintain in a desired low risk condition?

Scenario:

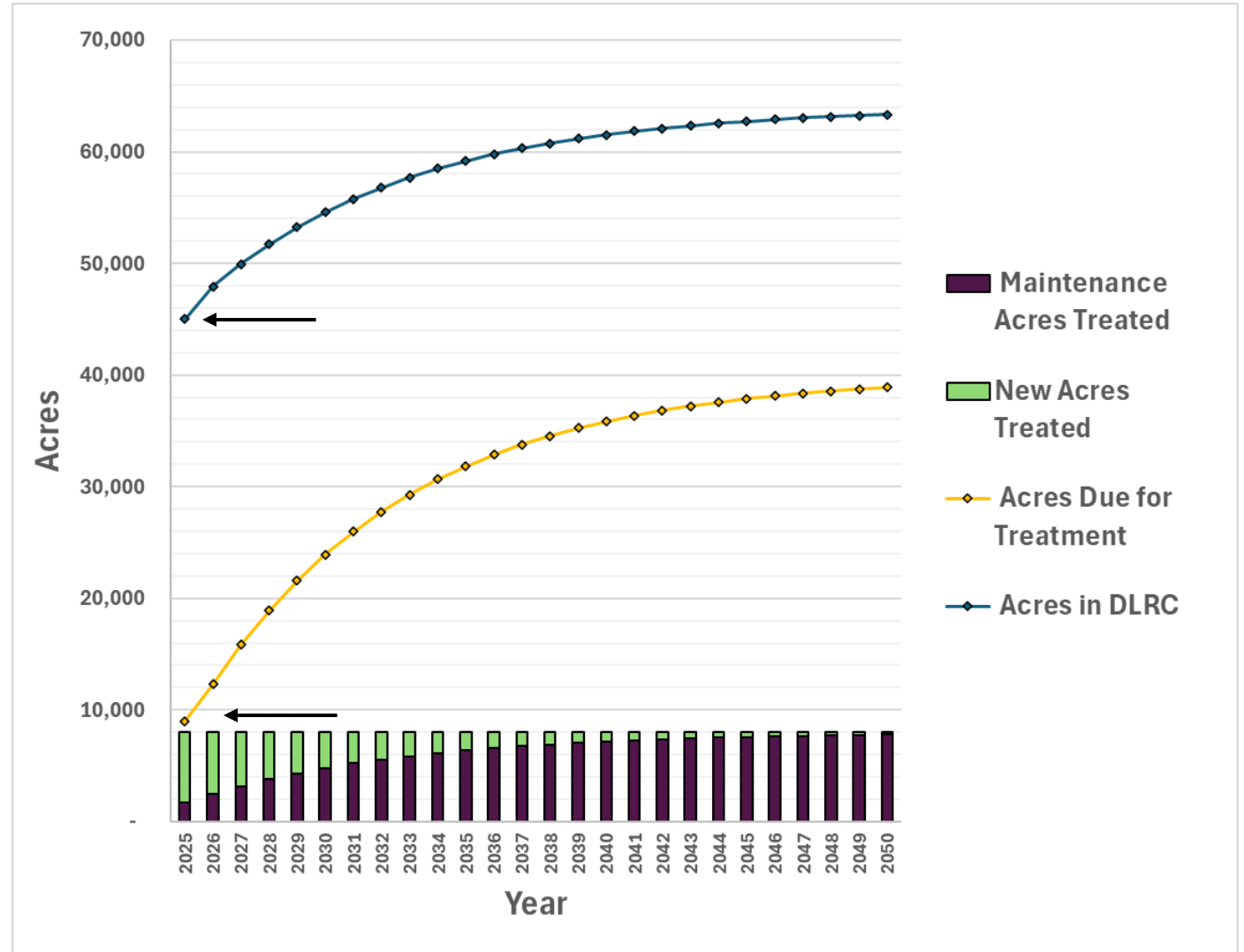
Bitterroot National Forest

- On average, treatments stay in DLRC for 8 years and return to pre-treatment hazard condition after 13 years.
- We expect 12% of DLRC to become Due each year, and 20% to go Out of Rotation each year.
- With that information, we can calculate the minimum number of acres we need to treat to maintain our previous investments.



How many acres can we maintain in a desired low risk condition?

In 2025, there were 45,000 acres in DLRC and 9,000 acres Due for Treatment.

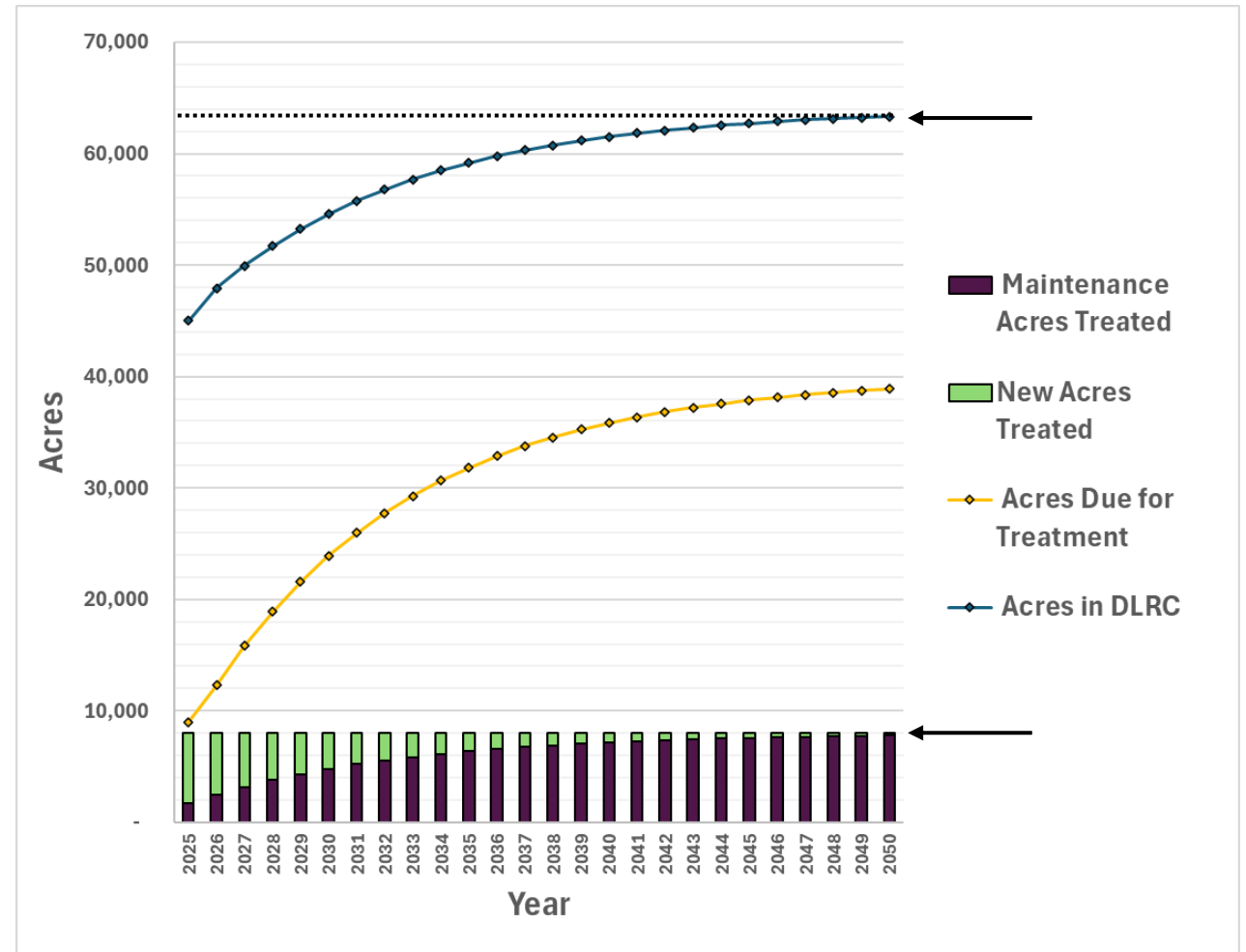


How many acres can we maintain in a desired low risk condition?

Eventually, we reach a point where all 8,000 acres need to be placed to maintain previous investments.

At this point, a threshold is reached for acres we can keep in DLRC.

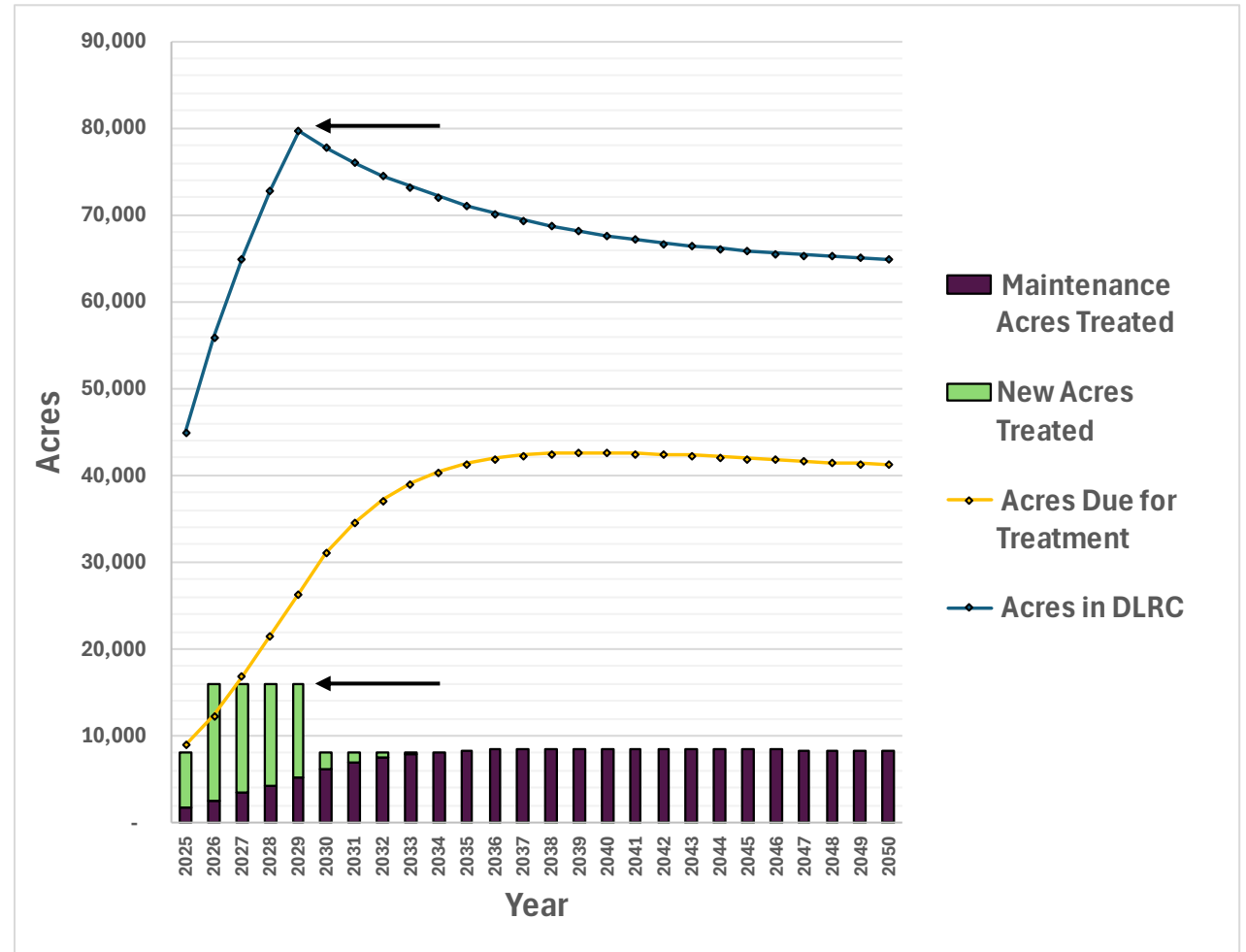
Given an **8,000-acre** annual treatment level, the maximum possible acres we can have in DLRC at any one time is **64,000 acres**.



How many acres can we maintain in a desired low risk condition?

Let's say we could double the amount of treatment from **8,000 acres** to **16,000 acres** for 4 consecutive years in 2026 to 2029.

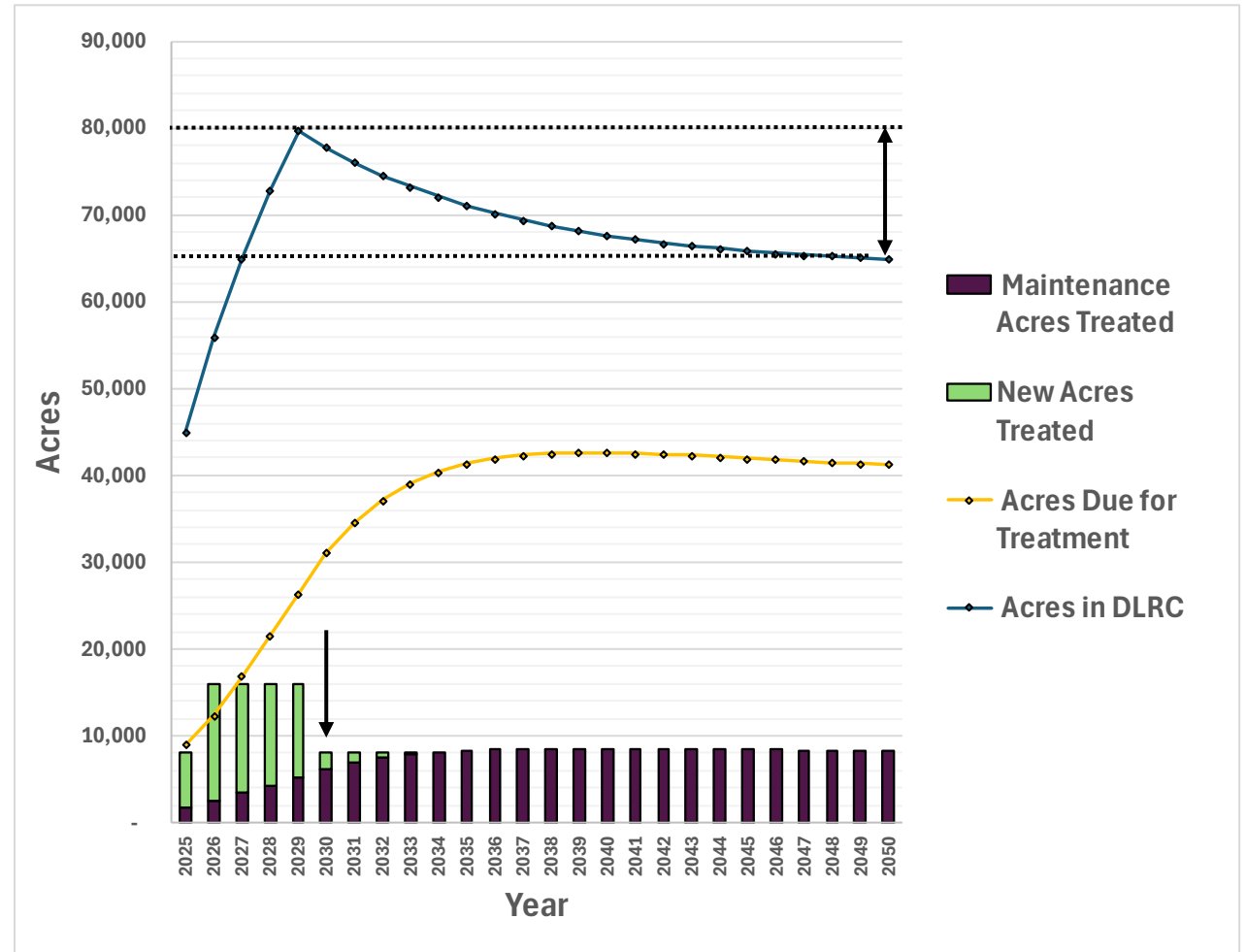
We reach a peak number of acres in DLRC very quickly. This peak is **80,000 acres** in DLRC in 2029.



How many acres can we maintain in a desired low risk condition?

When we return to the normal 8,000-acre treatment level in 2030, it is not enough to maintain the 4-year increase in treatment.

The number of acres in DLRC eventually decreases back to the 64,000-acre threshold. This means 16,000 treatment acres had a one-time effect on the landscape.



How many acres can we maintain in a desired low risk condition?

To ensure that the increased output in 2026 – 2029 does not fall out of rotation, the baseline number of acres treated starting in 2030 must **increase from 8,000 acres to 10,000 acres**.

By increasing acres treated moving forward, we can reach and maintain **80,000 acres in DLRC**.

